

Highlights

Lesion-Grounded Ordinal Grading with Clinical-Rule Consistency and Conformal Referral for Diabetic Retinopathy: Joint Classification, Lesion Segmentation and Lesion Detection on the DDR Dataset

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- Lesion evidence from a segmenter lifts DR grading kappa from 0.64 to 0.73.
- Whole-field lesion counts suffice; zone resolution and a clinical-rule loss added no measurable gain.
- Conformal referral sets meet nominal coverage only under exchangeable calibration.
- DDR validation and test partitions differ in difficulty; lesion and grading splits overlap.
- Whole study reproducible on a 4-core CPU; zero-shot external transfer failed and is reported.

Lesion-Grounded Ordinal Grading with Clinical-Rule Consistency and Conformal Referral for Diabetic Retinopathy: Joint Classification, Lesion Segmentation and Lesion Detection on the DDR Dataset

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Abstract

Background. Diabetic retinopathy (DR) is graded from lesions, yet grading, lesion segmentation and lesion detection are usually solved by separate models, and grades are not required to agree with lesion evidence. **Objective.** We test whether grounding the grade in lesion evidence from a segmentation network, with ordinal soft targets, a soft-logic clinical-rule loss and conformal referral sets, improves grading, and we report segmentation and detection on the same data. **Methods.** On DDR (13 673 fundus images, 757 with lesion masks and boxes), a U-Net yields four lesion maps that a differentiable bottleneck turns into 24 evidence tokens; a small transformer combines them with a frozen image embedding. All experiments ran on a four-core CPU; a leak-controlled test set (3633 images) was used because the lesion and grading partitions overlap. **Results.** Lesion evidence raised the quadratic-weighted kappa from 0.644–0.638 (embedding only) to 0.734 (95% CI 0.714–0.754; AUC for referable DR 0.923), above a fine-tuned ResNet-18 reference (0.700). Whole-field statistics carried most of this gain; zone resolution and the rule loss added nothing measurable, whereas soft targets lowered calibration error to 0.057. Segmentation reached a mean AUPR of 0.365 (microaneurysms ≤ 0.17) and detection a mean $AP_{0.3}$ of 0.168. Conformal sets reached nominal coverage under exchangeability but not across the official validation and test partitions, which differ in difficulty; zero-shot

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transfer to a 224×224 external set failed. **Conclusions.** Lesion-derived evidence is a cheap, clear source of grading information; other components were modest, and generalisation remains open.

Keywords: diabetic retinopathy, ordinal classification, lesion segmentation, lesion detection, clinical rule consistency, conformal prediction, fundus photography

1. Introduction

Diabetic retinopathy (DR) is a microvascular complication of diabetes mellitus and remains a leading cause of preventable vision impairment among working-age adults worldwide (Saeedi et al., 2019; Teo et al., 2021). A recent meta-analysis estimated that about one in five people living with diabetes (22.3%) has some degree of retinopathy, and projected that the absolute burden will keep increasing through the middle of this century as the prevalence of diabetes rises (Teo et al., 2021). Vision loss from DR is largely avoidable when disease is found early and treated on time, which is why periodic fundus photography is recommended for every person with diabetes. The bottleneck is not the camera but the reader: grading millions of photographs per year requires trained graders and ophthalmologists who are scarce in exactly those regions where diabetes is growing fastest. This has made automated DR screening one of the most extensively studied applications of deep learning in medical imaging, from the first large-scale validations (Gulshan et al., 2016; Ting et al., 2017) to prospective trials of autonomous systems in primary care (Abràmoff et al., 2018).

Clinically, DR is graded from a small set of visible lesions. Microaneurysms (MA), intraretinal haemorrhages (HE), hard exudates (EX) and soft exudates or cotton-wool spots (SE) are the hallmarks of non-proliferative disease, and neovascularisation marks the proliferative stage. The International Clinical Diabetic Retinopathy (ICDR) severity scale (Wilkinson et al., 2003), a simplification of the ETDRS classification (Early Treatment Diabetic Retinopathy Study Research Group, 1991), defines five grades — no DR, mild, moderate, severe non-proliferative DR and proliferative DR — by stating which lesions must be present and, for the severe stage, how widely they must be distributed (the well-known “4–2–1” rule: haemorrhages in four quadrants, venous beading in two, or intraretinal microvascular abnormalities in one). Three machine-learning tasks therefore describe the same clinical

reasoning at different granularity: *grading* (image level, five ordered classes), *lesion segmentation* (pixel level) and *lesion detection* (instance level).

In the literature, however, these three tasks are almost always solved separately. Grading networks are trained on image-level labels and are asked to rediscover lesions implicitly, which makes them data-hungry and hard to explain. Segmentation networks are trained on a few hundred pixel-annotated images and are scored by overlap measures that say little about whether a clinician could use the output. Detection networks are borrowed from natural-image object detection and struggle with objects that occupy a handful of pixels. Multi-task models that share an encoder do exist (Foo et al., 2020; Payout et al., 2019; Zhou et al., 2019), as do lesion-aware grading models (Huang et al., 2021; Sun et al., 2021; Wang et al., 2017), but in most of them the link between lesion evidence and the final grade is learned implicitly through shared weights. Nothing prevents such a model from predicting *severe* DR for an image in which it finds no haemorrhage at all, and the practitioner has no principled way to know when to distrust the output.

Gaps addressed in this work. We identify four specific gaps. **(G1) Lesion-grade inconsistency.** Grade predictions are not required to agree with the lesion evidence produced by the same system, so internally contradictory outputs are possible and are rarely measured. **(G2) Ordinal structure and label noise.** DR grades are ordered, and inter-grader disagreement concentrates on adjacent grades, yet cross-entropy over nominal classes ignores both facts. **(G3) Small-lesion sensitivity.** Microaneurysms are a few pixels wide in a megapixel image; losses and architectures that are fine for organs are blind to them. **(G4) Absence of principled referral.** Most systems output a single hard label with a miscalibrated confidence; very few provide a finite-sample statement about how often the clinically true grade is contained in the output, which is what a referral workflow needs.

Our approach. We propose LesionRule, a lesion-grounded grading framework built on top of a lesion-segmentation network. The segmentation network produces four lesion-probability maps. A differentiable, connected-component-free *lesion-evidence bottleneck* converts those maps into soft per-zone lesion statistics (counts at several confidence levels, number of local peaks and maximum probability for each of four lesion types in each of six retinal zones). Each (lesion, zone) statistic becomes a token; a small transformer lets a grade query attend jointly to a global image embedding and to these 24 evidence

tokens, so that the grade is computed *from* the lesions the same network sees. A *clinical-rule consistency loss* encodes ICDR-style necessary conditions (“grade $\geq$ severe requires haemorrhage in all four quadrants or heavy soft exudates”) as soft logic with a product t-norm, with thresholds calibrated on the training labels, and penalises predictions that exceed what the lesion evidence permits. The grade head is *ordinal* with Gaussian-smoothed soft targets that reflect adjacent-grade disagreement, and the resulting class probabilities are temperature-scaled and wrapped by *split-conformal prediction* to give referral sets with a finite-sample coverage guarantee. On the lesion side, a class-weighted overlap objective (compared in an ablation with plain cross-entropy and Dice) and a full-resolution shallow branch keep microaneurysms visible, and the very same probability maps are converted to detection boxes by connected-component proposals, so that segmentation and detection share one set of weights.

Experimental scope and honesty about compute. All experiments reported here were produced on a single four-core CPU workstation without a GPU (Section 5). This is a deliberate design constraint rather than an incidental one: it forces the study to be explicit about what is learned end-to-end (the lesion segmenter) and what is learned on top of frozen representations (the grading heads), and it makes all reported numbers reproducible without specialised hardware. It also means that absolute grading accuracy is below what an end-to-end fine-tuned, high-resolution, GPU-trained model could reach, and we do not claim state-of-the-art accuracy on the DDR benchmark. What we do claim, and test with controlled ablations, multiple seeds and confidence intervals, is that *given the same frozen image embedding*, grounding the grade in lesion evidence and enforcing rule consistency yields measurable gains in agreement, internal consistency, calibration and referral quality over the standard alternatives.

Contributions.

1. We formulate DR grading, lesion segmentation and lesion detection as a single coupled pipeline in which the grade is a function of differentiable, zone-wise lesion statistics (Section 4.5), and we make this coupling explicit with a token-based evidence head.
2. We introduce a soft-logic *clinical-rule consistency* regulariser with learnable, label-calibrated thresholds and a matching *rule-violation rate* met-

ric that quantifies internal contradictions between lesion evidence and predicted grade (Section 4.8).

3. We combine ordinal cumulative-link learning with Gaussian-smoothed soft targets and split-conformal prediction to obtain calibrated grade distributions and referral sets with finite-sample coverage (Sections 4.7 and 4.9).
4. We provide a lesion segmentation recipe (class-weighted objective, lesion-centred sampling, full-resolution branch), a controlled comparison of segmentation losses and input preprocessing, and a segmentation-derived detection protocol with FROC analysis for lesions that are only a few pixels wide, evaluated on the full DDR lesion subset (Section 4.3).
5. We report a comprehensive and fully reproducible empirical study on the DDR dataset (Li et al., 2019) — 13,673 fundus images, 757 of them with pixel masks and boxes — including loss and preprocessing ablations, component ablations over five seeds, calibration and conformal analyses, and a zero-shot external test on the Gaussian-filtered APTOS-derived Kaggle set (Asia Pacific Tele-Ophthalmology Society, 2019; Rath, 2020) (Section 6). All code and cached lesion maps are released.

Summary of findings. The results support some of these ideas and not others, and we say so at the outset. Lesion evidence from the segmentation network clearly improves grading over the same frozen image embedding (QWK $0.64 \rightarrow 0.73$) and over a fine-tuned small CNN; the soft ordinal targets improve calibration; and conformal sets deliver their guarantee under exchangeability. By contrast, the zone resolution of the evidence and the rule-consistency loss did *not* give measurable gains on DDR, the conformal guarantee fails across the official DDR validation and test partitions because they differ in difficulty, segmentation of microaneurysms and detection at strict overlap remain weak, and zero-shot transfer to a low-resolution external set fails. We regard these negative findings as part of the contribution, because they delimit what lesion-grounded grading can and cannot be expected to do with the resolution and data used here.

Organisation. Section 2 reviews related work and positions LesionRule against it. Section 3 describes the datasets and preprocessing. Section 4 details the method, with mathematical formulation and algorithms. Section 5 gives the experimental protocol. Section 6 presents results, and Section 7 discusses their clinical meaning, limitations and ethical aspects. Section 8 concludes.

2. Related work

2.1. Clinical background: lesions and grading scales

Diabetic retinopathy is a progressive microvascular disease. Chronic hyperglycaemia damages the capillary wall, producing the earliest visible change, the microaneurysm: a small, round, dark-red dot, a few tens of micrometres across, which is the first lesion on which a diagnosis of mild non-proliferative DR can be made. As capillaries leak or occlude, intraretinal haemorrhages (dot-and-blot or flame-shaped), hard exudates (yellowish lipid deposits, often arranged in rings around leaking vessels) and soft exudates or cotton-wool spots (retinal nerve-fibre-layer infarcts) appear. With progressive ischaemia, venous beading, intraretinal microvascular abnormalities and, finally, neovascularisation and pre-retinal or vitreous haemorrhage mark the transition to the proliferative stage, which carries a high risk of severe visual loss. Diabetic macular oedema, a separate complication that can occur at any stage, is the other main cause of vision loss and is graded in a different scale; it is not studied here.

The ICDR severity scale (Wilkinson et al., 2003) distinguishes five levels that can be assigned from a fundus photograph: no apparent retinopathy; mild non-proliferative DR (microaneurysms only); moderate non-proliferative DR (more than microaneurysms but less than severe); severe non-proliferative DR (any of: more than 20 intraretinal haemorrhages in each of four quadrants, definite venous beading in two or more quadrants, or prominent IRMA in at least one quadrant, the “4–2–1” rule); and proliferative DR (neovascularisation or vitreous or pre-retinal haemorrhage). Moderate or worse is conventionally called *referable* DR, and screening programmes refer such patients to an ophthalmologist; severe and proliferative disease are sight-threatening. The five levels are therefore not equally spaced clinically: the step from none to mild changes follow-up intervals, the step to moderate changes the referral decision, and the step to severe changes the urgency. This asymmetry matters for evaluation, since a metric such as accuracy or QWK weights all errors by distance on the grade axis, whereas a screening programme weights the *referral boundary* most; we therefore report AUC and referral sensitivity/specificity alongside QWK.

2.2. Deep learning for DR grading

The modern era of automated DR screening started with convolutional networks trained on large collections of graded photographs (Gulshan et al.,

2016), followed by multi-ethnic validations (Ting et al., 2017) and regulatory-grade prospective studies (Abràmoff et al., 2018). Subsequent research on public data has largely consisted of backbone and training-recipe improvements. Residual (He et al., 2016), compound-scaled (Tan and Le, 2019), modernised convolutional (Liu et al., 2022) and transformer backbones (Doso-vitskiy et al., 2021; Liu et al., 2021) have all been applied to grading, typically with high-resolution inputs, heavy augmentation and ordinal or regression ob-jectives. Because retinal images are structurally homogeneous and labelled data are scarce compared with natural images, large self-supervised retinal foundation models such as RETFound (Zhou et al., 2023) have recently shown that domain-specific pre-training transfers well across diseases and datasets. Our framework is complementary to this trend: it accepts any image em-bedding (here a frozen ImageNet ConvNeXt-T (Liu et al., 2022)) and adds lesion-grounded evidence and consistency on top.

A second line of work tackles the specific difficulties of DR grading. The class distribution is heavily skewed towards no-DR and moderate DR, and severe DR is rare. Category-attention modules (He et al., 2021) were pro-posed to counter this imbalance and were evaluated on DDR. Joint grading of DR and diabetic macular oedema exploits the strong relationship between the two diseases (Li et al., 2020). Noise-aware and attention-based fusion frameworks (Lin et al., 2018) address low-quality inputs. Common to these methods is that they supervise only the grade; the lesions that justify the grade are left implicit.

2.3. Lesion-aware grading and weak lesion localisation

Because lesions are the clinical basis of grading, a natural idea is to give the grader access to lesion information. Zoom-in-Net (Wang et al., 2017) mines suspicious regions with an attention branch trained only from image-level labels and re-examines them at higher resolution. Image-mining ap-proaches (Quellec et al., 2017) derive lesion heatmaps from a trained grading network. Two-stage methods first detect lesions and then feed lesion counts or maps to a grader (Yang et al., 2017). Lesion-aware transformers (Sun et al., 2021) cast lesion detection and grading in a joint transformer frame-work, and lesion-based contrastive learning (Huang et al., 2021) uses lesion patches to learn grading-relevant features. These works demonstrate that lesion information improves grading and interpretability. However, to our knowledge, they do not (i) make the grade a function of *zone-resolved* le-sion statistics with explicit tokens that can be inspected, (ii) check that the

final grade is *consistent* with the lesion evidence according to clinical-rule templates, or (iii) report a rule-violation measure, calibration and conformal coverage jointly. These are the aspects on which LesionRule differs.

2.4. Retinal lesion segmentation

Lesion segmentation in fundus images has been driven by encoder–decoder networks descending from U-Net (Ronneberger et al., 2015), including nested skip connections (Zhou et al., 2020), attention gates (Oktay et al., 2018) and atrous multi-scale decoders (Chen et al., 2018). Public pixel-level resources include IDRiD (Porwal et al., 2020), e-optha (Decencière et al., 2013), the larger FGADR collection (Zhou et al., 2021) and the DDR dataset used here (Li et al., 2019). The consistent finding across these benchmarks is that large, high-contrast lesions (hard exudates, large haemorrhages) are segmented reasonably well while microaneurysms remain difficult: they are small, low-contrast, often indistinguishable from vessel fragments, and their annotation is noisy. Overlap measures such as Dice are dominated by large lesions, whereas class-imbalance-robust measures such as the area under the precision–recall curve (AUPR) are more informative for rare, small objects (Saito and Rehmsmeier, 2015). Loss design is therefore important: Tversky-type losses (Salehi et al., 2017) and their focal variant (Abraham and Khan, 2019) weight false negatives more than false positives and are a standard remedy for tiny foreground classes, while focal modulation (Lin et al., 2017) down-weights easy background pixels. We adopt this family and add an explicit full-resolution branch, and we quantify the benefit through an ablation against binary cross-entropy and Dice-based objectives.

2.5. Lesion detection

Detection of retinal lesions has mostly reused generic object detectors (Ren et al., 2017; Tian et al., 2019). These architectures were designed for objects that span tens to hundreds of pixels, with anchors and strides that are poorly matched to lesions that may span only a few pixels, and they require many training boxes, while the DDR detection subset has only 383 training images. Detection from segmentation outputs is an attractive alternative when pixel masks are available: instances are obtained by connected-component analysis and ranked by their mean probability, and a single network serves both tasks. This is the route we take. Following common practice for tiny-object detection, we report average precision at relaxed and standard overlap thresholds (Padilla et al., 2021) together with lesion-level

free-response ROC (FROC) sensitivity at fixed numbers of false positives per image, which is the quantity that matters for screening workflows.

2.6. Multi-task learning and mixed supervision

Multi-task models for DR combine grading with lesion segmentation to exploit complementary supervision (Foo et al., 2020), learn lesion segmentation from image-level labels with weak supervision (Playout et al., 2019), or jointly train segmentation and classification in a semi-supervised, collaborative fashion (Zhou et al., 2019). Attention-based multiple-instance learning (Ilse et al., 2018) provides a general mechanism to localise image-level evidence. In all these designs, sharing is realised through common weights and a summed loss. Our contribution to this line is to make the dependence of the grade on the lesion maps *structural* rather than implicit, and to attach logic-based constraints to that dependence.

2.7. Ordinal learning, label noise and knowledge-based constraints

DR grades are ordered, so treating them as nominal categories discards information. Ordinal formulations convert a K -class problem to $K-1$ binary cumulative questions (Niu et al., 2016), with rank-consistent variants that guarantee monotone cumulative probabilities (Cao et al., 2020; Shi et al., 2023). Inter-grader disagreement in DR concentrates on adjacent grades, which suggests smoothing the target distribution along the grade axis; we adopt a discretised Gaussian target on the cumulative scale. Injecting logical knowledge into neural networks has a long history: logic rules can regularise network outputs through teacher–student distillation (Hu et al., 2016), semantic-based regularisation with t-norm relaxations (Diligenti et al., 2017), or semantic loss functions (Xu et al., 2018). We use the t-norm relaxation, because it is differentiable, requires no extra labels and gives an interpretable per-rule score. The ICDR/ETDRS grading definitions (Early Treatment Diabetic Retinopathy Study Research Group, 1991; Wilkinson et al., 2003) supply the templates. To our knowledge such rule templates have not previously been combined with zone-resolved lesion statistics from the same network and an explicit violation-rate metric in DR grading.

2.8. Calibration and conformal prediction

Modern networks are over-confident (Guo et al., 2017), and for screening the practically important question is what the probability that the true grade lies in the set of grades the system proposes is. Split-conformal prediction

(Angelopoulos and Bates, 2023; Vovk et al., 2005) converts any probabilistic classifier into a set-valued predictor with finite-sample marginal coverage under exchangeability, without retraining. Monte-Carlo dropout (Gal and Ghahramani, 2016) and deep ensembles estimate epistemic uncertainty but offer no coverage guarantee. We combine temperature scaling for calibration with split-conformal sets, define a clinically motivated referral rule on the resulting sets, and report coverage, set size and referral errors.

2.9. Positioning

Table 1 summarises how the methods discussed above differ in design. The comparison is qualitative: reported numbers of other works are obtained under different splits, resolutions, backbones and compute budgets, and we have not re-run those systems; therefore we make no numerical claims against them and restrict quantitative comparisons to controlled baselines trained under the same protocol (Section 5).

Table 1: Design-level comparison with representative prior work (qualitative; ✓ = addressed explicitly, ◦ = partially or implicitly, – = not addressed). Entries reflect our reading of the cited papers’ main design and may omit secondary features.

Method	Grade	Seg.	Det.	Lesion→grade link	Zone-wise evidence	Rule consist.	Ordinal / noise	Conformal
Zoom-in-Net (Wang et al., 2017)	✓	–	◦	◦	–	–	–	–
Two-stage detect. + grade (Yang et al., 2017)	✓	–	✓	✓	–	–	–	–
Foo et al. (Foo et al., 2020)	✓	✓	–	◦	–	–	–	–
Playout et al. (Playout et al., 2019)	–	✓	–	–	–	–	–	–
CANet (Li et al., 2020)	✓	–	–	–	–	–	–	–
CABNet (He et al., 2021)	✓	–	–	–	–	–	◦	–
Lesion-aware transf. (Sun et al., 2021)	✓	–	✓	✓	–	–	–	–
Lesion-based contrastive (Huang et al., 2021)	✓	–	–	◦	–	–	–	–
LesionRule (this work)	✓	✓	✓	✓	✓	✓	✓	✓

3. Materials

3.1. The DDR dataset

All primary experiments use the DDR dataset (also distributed as OIA-DDR) (Li et al., 2019), a general-purpose collection of colour fundus photographs designed for three tasks: DR grading, lesion segmentation and lesion detection. According to the dataset authors, the photographs were collected from many hospitals and acquired with a variety of fundus cameras, which

gives the data heterogeneous colour, illumination and field-of-view characteristics (Fig. 1). We downloaded the complete archive (ten split zip volumes, 15.1 GB) and verified its structure and counts directly from the files: the grading task contains 13 673 images split into 6835 training, 2733 validation and 4105 test images; the lesion subset contains 757 images (383 / 149 / 225) with four-class pixel-level masks and bounding boxes in Pascal-VOC XML format. The dataset is released under CC BY-NC-SA 4.0, and we use it for non-commercial research with attribution as required.

Grading labels. Each image carries one of six labels: grades 0 (no DR), 1 (mild), 2 (moderate), 3 (severe), 4 (proliferative DR) and 5 (ungradable owing to poor quality). The label distribution is strongly imbalanced (Table 2, Fig. 3a): 46% of training images are grade 0 and 33% grade 2, whereas severe DR (grade 3) accounts for only 118 training images (1.7%). The distribution is nearly identical across the three splits, which suggests stratified assignment. Throughout the paper we use two protocols. The *five-class protocol* (main) restricts grading to gradable images (grades 0–4: 6260 / 2503 / 3759 train / validation / test images), because ordinal reasoning is meaningless for an ungradable photograph; image quality is handled by a separate gate (Section 4.10). The *six-class protocol* adds the ungradable class as a nominal outcome and is reported for completeness.

Lesion annotations. For the 757 lesion images, four lesion types are annotated at pixel level: microaneurysms (MA), haemorrhages (HE), hard exudates (EX) and soft exudates (SE). The annotation is dense (Fig. 2): after removing duplicates, the training subset contains 24 671 annotated boxes, of which 13 419 are exudates, 5783 microaneurysms, 4739 haemorrhages and 730 soft exudates (the raw training XML files list 58 200 boxes, because many boxes appear more than once; the validation and test files contain no duplicates). Lesion size spans two orders of magnitude (Fig. 3c): when sizes are rescaled to the nominal ≈ 1950 -pixel native resolution, the median longer box side is about 10 px for microaneurysms and exudates, 26 px for haemorrhages and 47 px for soft exudates, and the 10th percentile is only 3.6–4.7 px for MA, EX and SE. In the resolution used by most deep networks (512 px or lower), a typical microaneurysm therefore occupies one to three pixels. The detection annotations are derived from the same lesions as the masks. Table 2 also lists the number of lesion images in which each class occurs.

Table 2: DDR dataset statistics as verified from the downloaded archive. Top: grading labels per split (G0–G4 = ICDR grades, UG = ungradable). Bottom: lesion subset (unique boxes after duplicate removal; number of images containing each lesion class in parentheses).

Split	G0	G1	G2	G3	G4	UG	Total
Train	3,133	315	2,238	118	456	575	6,835
Validation	1,253	126	895	47	182	230	2,733
Test	1,880	189	1,344	71	275	346	4,105
All	6,266	630	4,477	236	913	1,151	13,673

Lesion split	Images	MA	HE	EX	SE
Train	383	5,783 (314)	4,739 (294)	13,419 (245)	730 (111)
Validation	149	2,556 (132)	1,342 (113)	1,920 (70)	349 (86)
Test	225	2,041 (124)	6,457 (194)	8,320 (171)	214 (42)

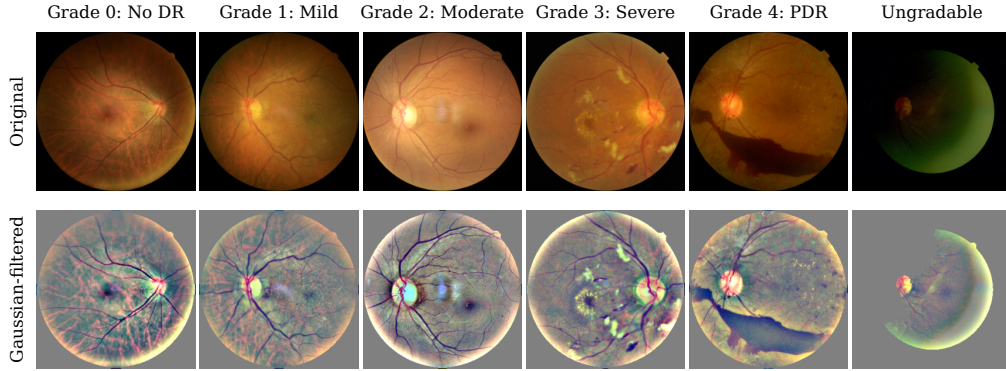


Figure 1: Sample DDR fundus images for each grade (top row: cropped original; bottom row: after Gaussian-filter preprocessing). Samples are drawn at random from the training split with a fixed seed, subject to the field of view filling the frame; the last column shows an ungradable image. The preprocessing equalises illumination and exposes vessels and lesions, but amplifies border artefacts.

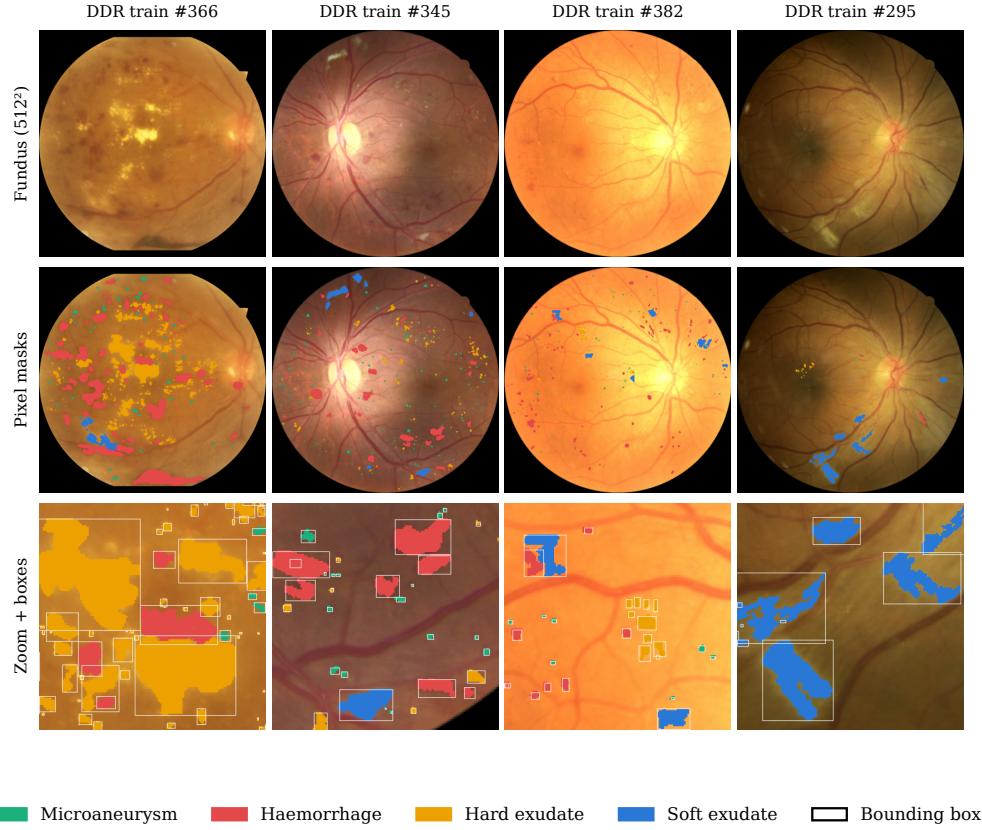


Figure 2: Lesion annotations in four DDR training images that contain several lesion types. Top: fundus image at 512×512 . Middle: ground-truth pixel masks (MA green, HE red, EX yellow, SE blue). Bottom: densest 128×128 window magnified three times with the Pascal-VOC bounding boxes (white). Note the very different sizes and densities of lesions between images.

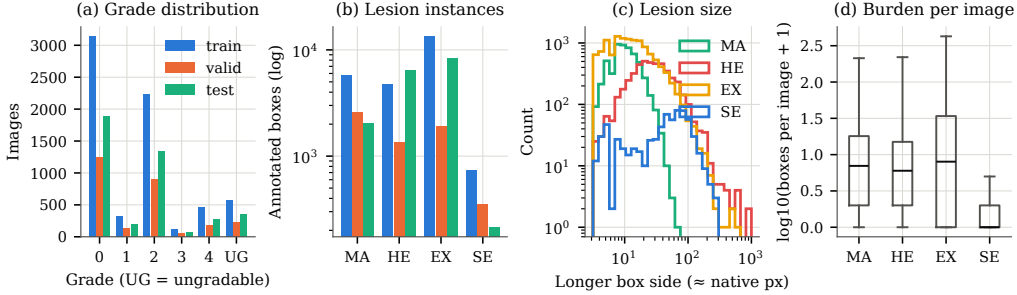


Figure 3: Dataset statistics. (a) Grade distribution per split. (b) Number of annotated lesion boxes per class and split (log scale). (c) Distribution of the longer side of lesion boxes in the training set, in approximate native pixels. (d) Lesion burden per image (boxes per image, $\log_{10}$ scale, outliers omitted).

3.2. Gaussian-filtered APTOS-derived Kaggle set (external test)

For an external test of the grading pipeline we use the publicly available “Diabetic Retinopathy 224×224 Gaussian filtered” Kaggle dataset (Rath, 2020), which contains the labelled training portion of the APTOS 2019 Blindness Detection challenge (Asia Pacific Tele-Ophthalmology Society, 2019) after resizing to 224×224 and Gaussian-filter normalisation in the style of the Kaggle 2015 winning solution (Graham, 2015). It contains 3664 PNG images in five folders (No_DR 1805, Mild 370, Moderate 999, Severe 193, Proliferative_DR 295; CC0 licence). This set differs from DDR in several ways that make it a demanding test of robustness: it originates from a different population and camera set (Aravind Eye Hospital, India), has a different grade distribution (49% grade 0, 27% grade 2), has no lesion annotation and, most importantly, has a resolution of only 224×224 , which removes microaneurysms and most small haemorrhages (Fig. 4). We therefore *only* use it for zero-shot grading evaluation of models trained on DDR, and not for segmentation or detection. Because the images are already filtered, we apply the same filter to DDR inputs wherever a model is to be transferred (Section 3.3).

3.3. Preprocessing

All photographs are processed as follows. (i) *Border cropping*: the bounding box of pixels whose maximum channel exceeds 12 is cropped and the result is zero-padded to a square, which removes the black margins that occupy a variable portion of DDR frames. (ii) *Resizing*: area interpolation

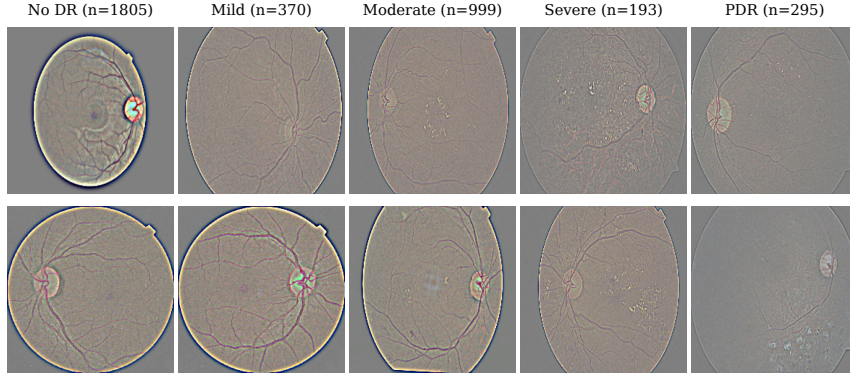


Figure 4: Random samples (two per grade) from the Gaussian-filtered 224×224 APTOS-derived Kaggle set. The Gaussian filter equalises illumination but also removes most colour information; note that some photographs are non-circular owing to the original acquisition.

to 512×512 for the segmentation network and to 256×256 for the image embedding. Binary lesion masks are resized with area interpolation of the floating-point mask followed by a threshold of 0.15, which preserves sub-pixel microaneurysms that nearest-neighbour or threshold-at-0.5 resizing would erase; ground-truth boxes undergo exactly the same geometric transform. (iii) *Gaussian filtering* (optional, denoted “GF”): $I' = 4I - 4(G_\sigma * I) + 128$ with $\sigma = 0.045s$ (where s is the image side, matching the common $\sigma \approx 10$ at 224 px), followed by a gray fill outside the field of view. This is the same transform used to produce the Kaggle set, and we study its effect in an ablation (Section 6.1).

3.4. Data splits, leakage and ethics

We use the official DDR train/validation/test partition for every task. Model selection (early stopping, thresholds, temperature, conformal calibration) uses validation data only. A subtle point is that the lesion subset is *not* aligned with the grading partition: of the 383 lesion-training images, 275, 100 and 8 belong to the grading training, validation and test partitions, respectively; of the 149 lesion-validation images, 31 are in the grading validation and 118 in the grading test partition; and of the 225 lesion-test images, 152, 12 and 59 are in the grading training, validation and test partitions (Appendix Appendix A). Because the segmentation network is fitted on lesion-training images and tuned on lesion-validation images, any grading-test image that belongs to either set would give the lesion branch indirect

access to test data. We therefore define a *leak-controlled grading test set* by removing these 126 images (all of which are gradable, grades 1–4), leaving 3633 gradable test images, and report *all* grading test results on this set; results on the full official test partition are given as a sensitivity analysis in Appendix Appendix A. Lesion-test images are never used for fitting or selection of the segmentation network, so segmentation and detection results are unaffected. Both datasets are public and de-identified; no new patient data were collected, so institutional review was not required.

4. Method

4.1. Problem formulation and notation

Let $x \in \mathbb{R}^{H \times W \times 3}$ be a fundus photograph. We consider three coupled tasks on the same input. *Grading* asks for an ordinal grade $y \in \{0, 1, 2, 3, 4\}$ (Section 3.1). *Segmentation* asks for four binary masks $M_c \in \{0, 1\}^{H \times W}$ for lesion classes $c \in \mathcal{C} = \{\text{MA}, \text{HE}, \text{EX}, \text{SE}\}$. *Detection* asks for a set of scored boxes $\{(b_j, c_j, s_j)\}$. A segmentation network f_ϕ maps the image to lesion probability maps $P = f_\phi(x) \in [0, 1]^{4 \times H \times W}$; a grading model g_ψ maps the image and the lesion evidence derived from P to cumulative grade probabilities. Table 3 lists the main symbols.

4.2. Overview

Fig. 5 shows the overall pipeline. The segmentation network is trained on the pixel-annotated DDR subset and applied to every image. Its probability maps are summarised by the lesion-evidence bottleneck into a tensor Z of soft, zone-wise lesion statistics (Section 4.5). The grading head combines a frozen embedding of the same image with the evidence tokens through a transformer (Section 4.6) and outputs cumulative ordinal probabilities (Section 4.7). A rule module computes ICDR-style necessary-condition scores $R_k(Z)$ and penalises predictions that exceed them (Section 4.8). Probabilities are temperature-scaled and converted into conformal prediction sets (Section 4.9). Detection boxes are obtained from the same probability maps P (Section 4.4).

4.3. Lesion segmentation network

Architecture. The segmenter f_ϕ is a U-Net (Ronneberger et al., 2015) with a ResNet-18 encoder (He et al., 2016) initialised from ImageNet weights (as distributed in `timm` (Wightman, 2019)). The encoder produces features at

Table 3: Main notation.

Symbol	Meaning
x, y	fundus image, ICDR grade $\in \{0, \dots, 4\}$
$\mathcal{C} = \{\text{MA}, \text{HE}, \text{EX}, \text{SE}\}$	lesion classes
$P = f_\phi(x) \in [0, 1]^{4 \times H \times W}$	lesion probability maps from the segmentation network
$\tilde{P} \in [0, 1]^{4 \times h \times w}$	$4 \times$ max-pooled maps ($h = w = 128$)
$\mathcal{Z} = \{z_1, \dots, z_6\}, \Omega_z$	retinal zones (4 quadrants, central disc, whole field) and their pixel sets
$n_{c,z}^{(\tau)}, \pi_{c,z}, \mu_{c,z}$	soft counts at threshold τ , number of peaks, maximum probability
$e_{c,z} \in \mathbb{R}^5, Z \in \mathbb{R}^{4 \times 6 \times 5}$	evidence vector of (c, z) and the full evidence tensor
$\varphi(x) \in \mathbb{R}^D$	frozen image embedding ($D = 1152$)
$s_k, \hat{F}_k = \sigma(s_k)$	cumulative logit and probability for the event $y \geq k$, $k = 1, \dots, 4$
$R_k(Z; \theta) \in [0, 1]$	soft-logic rule score: necessary condition for $y \geq k$
σ_y, λ	soft-label width, rule-loss weight
$\mathcal{C}_\alpha(x)$	conformal prediction set at miscoverage level α

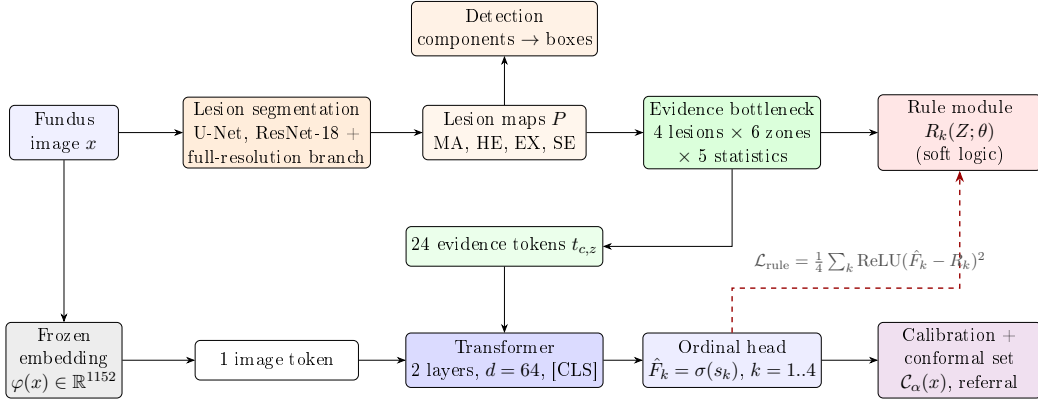


Figure 5: Overview of LesionRule. A segmentation network produces lesion maps; a differentiable bottleneck turns them into zone-wise evidence tokens, which a transformer combines with a frozen image embedding to predict cumulative ordinal probabilities. A soft-logic rule module (dashed) penalises grades that the lesion evidence cannot justify. Probabilities are calibrated and wrapped into conformal sets. The same lesion maps yield detection boxes.

strides 2, 4, 8, 16 and 32; the decoder upsamples bilinearly and fuses encoder features by concatenation followed by two 3×3 convolution–BN–ReLU blocks (widths 128, 96, 64, 48). Because micro-aneurysms are only a few pixels wide, stride-2 features are not sufficient, and we add a shallow *full-resolution branch* (two 3×3 conv–BN–ReLU blocks with 16 channels applied directly to the input) whose output is concatenated with the upsampled decoder output before a final 3×3 block and a 1×1 convolution to four logits. The network has 11.8 M parameters.

Objective. Let $p_c = \sigma(\ell_c)$ be the predicted probability map and y_c the binary mask of class c . We compare three objectives:

$$\mathcal{L}_{\text{bce}} = \frac{1}{4} \sum_c \text{BCE}(p_c, y_c), \quad (1)$$

$$\mathcal{L}_{\text{bce+dice}} = \frac{1}{4} \sum_c [\text{BCE}_w(p_c, y_c) + \mathcal{L}_{\text{Dice}}(p_c, y_c)], \quad (2)$$

$$\mathcal{L}_{\text{seg}} = \frac{1}{4} \sum_c [\text{BCE}_w(p_c, y_c) + 2 \text{FTL}_c], \quad (3)$$

where BCE_w weights positive pixels by $w = (20, 6, 6, 8)$ for (MA, HE, EX, SE), reflecting that the smaller and rarer the lesion class, the stronger the class imbalance. $\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum p y + 1}{\sum p + \sum y + 1}$, and the focal Tversky loss (Abraham and Khan, 2019; Salehi et al., 2017) is

$$\text{FTL}_c = \left(1 - \text{TI}_c\right)^\gamma, \quad \text{TI}_c = \frac{\text{TP}_c + 1}{\text{TP}_c + \alpha \text{FP}_c + \beta \text{FN}_c + 1}, \quad (4)$$

with $\text{TP}_c = \sum p_c y_c$, $\text{FP}_c = \sum p_c (1 - y_c)$, $\text{FN}_c = \sum (1 - p_c) y_c$ computed over the mini-batch, $\alpha = 0.3$, $\beta = 0.7$ and $\gamma = 0.75$. Since $\beta > \alpha$, missed lesion pixels cost more than spurious ones, which is the desirable bias for screening where recall of small lesions dominates; $\gamma < 1$ gives larger gradients to well-classified classes than the plain Tversky loss, and keeps the loss sensitive when overlap is already moderate. $\mathcal{L}_{\text{seg}}$ is the default objective used below; $\mathcal{L}_{\text{bce}}$ and $\mathcal{L}_{\text{bce+dice}}$ are the ablation baselines. We make no a-priori claim that the focal Tversky objective is superior to Dice: Section 6.1 reports the comparison.

Lesion-centred sampling and optimisation. Positive pixels are less than 1% of all pixels. Each training crop (256×256) is, with probability 0.7, positioned around a pixel of a randomly chosen lesion class in an image containing that class (the offset of the lesion inside the crop is random), and with probability 0.3 sampled uniformly. We use random flips, 90° rotations

and brightness/contrast jitter. Optimisation uses AdamW (Loshchilov and Hutter, 2019) with weight decay 10^{-4} , one-cycle learning-rate schedule (peak 10^{-3} , 15% warm-up), batches of 8 crops, 48 iterations per epoch, 30 epochs and bfloat16 autocast on CPU (Alg. 1). The checkpoint with the highest mean AUPR on the lesion-validation split (evaluated every five epochs at the full 512×512 resolution) is retained.

Algorithm 1 Lesion-centred training of the segmentation network.

Require: lesion-train images $\{(x_i, Y_i)\}$, epochs E , iterations I , crop size q , probability $\rho = 0.7$

- 1: **for** $e = 1, \dots, E$ **do**
- 2: **for** $t = 1, \dots, I$ **do**
- 3: **for** $b = 1, \dots, 8$ **do**
- 4: draw $u \sim \mathcal{U}(0, 1)$
- 5: **if** $u < \rho$ **then**
- 6: draw class c , image i with $Y_{i,c} \neq 0$, lesion pixel r ; place crop containing r at a random offset
- 7: **else**
- 8: draw image i and crop position uniformly
- 9: **end if**
- 10: augment (flip, rot90°, brightness/contrast) crop and mask
- 11: **end for**
- 12: $\ell \leftarrow f_\phi(\text{batch})$ (bf16); $\mathcal{L} \leftarrow \mathcal{L}_{\text{seg}}(\ell, \text{masks})$ with Eq. (4)
- 13: update ϕ with AdamW and one-cycle schedule
- 14: **end for**
- 15: **if** $e \bmod 5 = 0$ **then**
- 16: evaluate mean AUPR on lesion-valid at 512^2 ; keep best ϕ
- 17: **end if**
- 18: **end for**

4.4. Detection from segmentation maps

For a class c and image, the probability map P_c is thresholded at a class-specific seed threshold ϑ_c . Each 8-connected component K of $\{u : P_c(u) \geq \vartheta_c\}$ becomes a box b_K (its tight bounding rectangle) with score $s_K = \frac{1}{|K|} \sum_{u \in K} P_c(u)$. Because the segmentation objective is trained with positive-class weights, probabilities are not calibrated and the Dice-optimal threshold is high (typically > 0.8); using it as seed would cap the attainable

recall. We therefore choose ϑ_c from the grid $\Theta = \{0.05, 0.1, 0.2, 0.3, 0.5, 0.7\}$ by maximising AP at IoU 0.3 on the lesion-*validation* split. Lower seeds have higher recall but more false positives; the scores order the boxes for precision–recall and FROC analysis (Alg. 2). No extra network is trained: segmentation and detection share the same weights, and we evaluate the proposals directly against the DDR ground-truth boxes.

Algorithm 2 Segmentation-guided lesion proposals and evaluation.

Require: probability maps $P \in [0, 1]^{4 \times H \times W}$, seed grid Θ , validation split with ground-truth boxes

- 1: **for** each class $c \in \mathcal{C}$ **do**
- 2: $\vartheta_c \leftarrow \arg \max_{\vartheta \in \Theta} \text{AP}_{0.3}^{\text{val}}(\vartheta)$ (computed once on validation images);
 $B_c \leftarrow \emptyset$
- 3: find 8-connected components $\{K\}$ of $\{u : P_c(u) \geq \vartheta_c\}$
- 4: **for** each component K **do**
- 5: $b_K \leftarrow$ bounding box of K ; $s_K \leftarrow \text{mean}_{u \in K} P_c(u)$; add (b_K, s_K) to B_c
- 6: **end for**
- 7: **end for**
- 8: **evaluate:** sort B_c by score; greedy one-to-one matching to unique ground-truth boxes with $\text{IoU} \geq \kappa$ (AP_κ , $\kappa \in \{0.3, 0.5\}$); lesion-level FROC: a box is a hit if its centre lies inside a ground-truth box

4.5. Lesion-evidence bottleneck

The grading head does not see P directly; it sees a compact, interpretable summary computed as follows.

Zones. Let $h = w = 128$ and let $\tilde{P} = \text{maxpool}_4(P) \in [0, 1]^{4 \times h \times w}$. Max-pooling (rather than average pooling) keeps the peak of a one-pixel microaneurysm. We define six zones with binary masks Ω_z on the $h \times w$ grid, restricted to the circular field of view of radius $0.5h - 2$: four quadrants (split by the horizontal and vertical lines through the image centre), a central disc of radius $0.35 \cdot (h/2)$ approximating the macular region, and the whole field. Because the cropped photographs are approximately centred on the optic disc–macula axis, quadrants are a pragmatic stand-in for the ETDRS/ICDR quadrant definition; we do not claim anatomical registration.

Soft counts. Counting connected components is not differentiable and is unstable under thresholding. Instead, for threshold levels $\tau \in \{0.25, 0.5, 0.75\}$ we compute

$$n_{c,z}^{(\tau)} = \sum_{u \in \Omega_z} \sigma\left(\frac{\tilde{P}_c(u) - \tau}{T}\right), \quad T = 0.05, \quad (5)$$

which counts the number of pooled cells in zone z that contain lesion class c with confidence about τ . Two further statistics are the number of local peaks $\pi_{c,z} = \sum_{u \in \Omega_z} \mathbb{1}[\tilde{P}_c(u) = \max_{u' \in \mathcal{N}_3(u)} \tilde{P}_c(u') \wedge \tilde{P}_c(u) > 0.5]$ (a proxy for the lesion count) and the maximum probability $\mu_{c,z} = \max_{u \in \Omega_z} \tilde{P}_c(u)$. The evidence vector is

$$e_{c,z} = \log(1 + [n_{c,z}^{(0.25)}, n_{c,z}^{(0.5)}, n_{c,z}^{(0.75)}, \pi_{c,z}, 10\mu_{c,z}]) \in \mathbb{R}^5, \quad (6)$$

and the tensor $Z = (e_{c,z}) \in \mathbb{R}^{4 \times 6 \times 5}$ (120 numbers) is the *only* way in which lesion information enters the grade (Alg. 3). The logarithm compresses the dynamic range between a handful of microaneurysms and several hundred exudate cells. Since Eq. (5) is smooth in $\tilde{P}$, the bottleneck is differentiable in principle, which would allow end-to-end fine-tuning of f_ϕ from grade supervision; in the present study, f_ϕ is frozen at this stage for compute reasons (Section 5).

Algorithm 3 Zone-wise soft lesion evidence.

Require: lesion probability maps $P \in [0, 1]^{4 \times H \times W}$, zone masks $\{\Omega_z\}_{z=1}^6$ on the 128^2 grid

```

1:  $\tilde{P} \leftarrow \text{maxpool}_4(P)$ 
2: for  $c \in \mathcal{C}$ ,  $z = 1..6$  do
3:   for  $\tau \in \{0.25, 0.5, 0.75\}$  do
4:      $n_{c,z}^{(\tau)} \leftarrow \sum_{u \in \Omega_z} \sigma((\tilde{P}_c(u) - \tau)/T)$ 
5:   end for
6:    $\pi_{c,z} \leftarrow \sum_{u \in \Omega_z} \mathbb{1}[\tilde{P}_c(u) \text{ is a } 3 \times 3 \text{ local maximum, } \tilde{P}_c(u) > 0.5]$ 
7:    $\mu_{c,z} \leftarrow \max_{u \in \Omega_z} \tilde{P}_c(u)$ 
8:    $e_{c,z} \leftarrow \log(1 + [n_{c,z}^{(0.25)}, n_{c,z}^{(0.5)}, n_{c,z}^{(0.75)}, \pi_{c,z}, 10\mu_{c,z}])$ 
9: end for
10: return  $Z = (e_{c,z})$ 

```

4.6. Evidence-token grade head

Each (c, z) pair becomes a token $t_{c,z} = W_e e_{c,z} + a_c + b_z \in \mathbb{R}^d$ with $d = 64$, where a_c and b_z are learned class and zone embeddings. The frozen image embedding $\varphi(x) \in \mathbb{R}^{1152}$ (concatenated global averages of stages 3 and 4 of a ConvNeXt-T (Liu et al., 2022) applied to the 256×256 filtered image, standardised with training-set statistics) becomes one image token $t_0 = W_i \text{Dropout}(\varphi(x))$. A learned [CLS] token and the $1 + 24$ tokens enter a two-layer pre-LN transformer encoder (four heads, feed-forward width 128, dropout 0.1). The output of the [CLS] token passes through LayerNorm and a linear layer to produce four cumulative logits $s_1, \dots, s_4$. Attention weights from the [CLS] token to the evidence tokens provide a direct, inspectable account of which lesion class and zone drove a prediction. The head has 0.14 M parameters, so that the whole grading stage can be trained in seconds on a CPU.

4.7. Ordinal cumulative-link objective with soft targets

Following the multiple-output formulation of ordinal regression (Cao et al., 2020; Niu et al., 2016), we model $\hat{F}_k = P(y \geq k \mid x) = \sigma(s_k)$ for $k = 1, \dots, 4$. To model adjacent-grade disagreement we replace the one-hot target by a discretised Gaussian over grades,

$$q_j(y) = \frac{\exp(-(j-y)^2/2\sigma_y^2)}{\sum_{j'=0}^4 \exp(-(j'-y)^2/2\sigma_y^2)}, \quad F_k^* = \sum_{j \geq k} q_j(y), \quad (7)$$

and minimise the binary cross-entropy between F_k^* and $\hat{F}_k$:

$$\mathcal{L}_{\text{ord}} = -\frac{1}{4} \sum_{k=1}^4 \left[F_k^* \log \hat{F}_k + (1 - F_k^*) \log(1 - \hat{F}_k) \right]. \quad (8)$$

With $\sigma_y = 0$ this reduces to the standard cumulative-link loss. At inference, $\hat{y} = \sum_k \mathbb{I}[\hat{F}_k > 0.5]$, and the class probabilities are obtained from the monotonised cumulative probabilities $\bar{F}_k = \min_{k' \leq k} \hat{F}_{k'}$ as $p_j = \bar{F}_j - \bar{F}_{j+1}$ (with $\bar{F}_0 = 1$, $\bar{F}_5 = 0$), floored at 10^{-6} and renormalised. In our experiments, $\sigma_y = 0.5$, which places about 0.79 of the target mass on the labelled grade and 0.10 on each neighbour.

4.8. Clinical-rule consistency

The ICDR scale describes grades through necessary conditions on the lesions that must be visible. We encode four such conditions as soft logic (Diligenti et al., 2017; Hu et al., 2016), using the product t-norm for conjunction ($a \wedge b = ab$) and its dual for disjunction ($a \vee b = 1 - (1 - a)(1 - b)$). Let $u_{c,z} = e_{c,z,2} = \log(1 + n_{c,z}^{(0.5)})$ and $a(v; \theta) = \sigma((v - \theta)/s)$ with $s = 0.35$. With learnable thresholds $\theta = (\theta_1, \dots, \theta_{10})$ and $z = \text{all the whole-field zone}$, the rule scores are

$$R_1 = \bigvee_{c \in \mathcal{C}} a(u_{c,\text{all}}; \theta_c), \quad (9)$$

$$R_2 = a(u_{\text{MA},\text{all}}; \theta_5) \vee a(u_{\text{HE},\text{all}}; \theta_2) \vee a(u_{\text{EX},\text{all}}; \theta_3) \vee a(u_{\text{SE},\text{all}}; \theta_4), \quad (10)$$

$$R_3 = \left(\bigwedge_{q=1}^4 a(u_{\text{HE},q}; \theta_{5+q}) \right) \vee a(u_{\text{SE},\text{all}}; \theta_{10}), \quad (11)$$

$$R_4 = R_2, \quad (12)$$

where θ_c in Eq. (9) denotes the thresholds $\theta_1, \dots, \theta_4$ for $c = \text{MA, HE, EX, SE}$. The four rules read as follows. R_1 : some lesion is visible (needed for mild DR or worse). R_2 : lesions beyond a few microaneurysms are visible (needed for moderate or worse). R_3 : haemorrhages in all four quadrants, *or* extensive soft exudates (a soft proxy for the “4” of the 4–2–1 rule; needed for severe or worse). R_4 : proliferative DR is never lesion-free. Venous beading, IRMA and neovascularisation are not annotated in DDR, so R_3 is only a partial surrogate and R_4 imposes just a weak necessary condition. We stress that these rules are *necessary conditions* (“ $y \geq k \Rightarrow R_k$ ”), not sufficient ones: a grade may be lower than the lesions allow, but never higher.

Calibrating thresholds on labels. Because the predicted lesion statistics are noisy, hand-set thresholds would be arbitrary. We therefore fit θ once on the training labels by minimising

$$\mathcal{L}_{\text{fit}}(\theta) = \sum_{k=1}^4 \left[\mathbb{E}_{i: y_i \geq k} (1 - R_k(Z_i; \theta)) + \frac{1}{2} \mathbb{E}_{i: y_i < k} R_k(Z_i; \theta) \right], \quad (13)$$

that is, the rule should hold for images whose grade is at least k (first term), and should not hold trivially for lesion-poor images (second term, which prevents $R_k \equiv 1$). The parameters are then frozen. The rules are thus *clinically inspired, data-calibrated templates*, not a re-implementation of the grading guideline.

Rule-consistency loss and violation rate. The grading head is trained with

$$\mathcal{L} = \mathcal{L}_{\text{ord}} + \lambda \mathcal{L}_{\text{rule}}, \quad \mathcal{L}_{\text{rule}} = \frac{1}{4} \sum_{k=1}^4 \text{ReLU}(\hat{F}_k - R_k(Z; \theta))^2, \quad \lambda = 1, \quad (14)$$

so that the predicted probability of grade $\geq k$ is pushed below the rule score. As an evaluation measure independent of the training loss, we define the *rule-violation rate* of a model $\hat{y}(\cdot)$ on a test set as the fraction of images for which $\hat{y} \geq k$ and $R_k < 0.5$ for at least one $k \leq \hat{y}$ (Alg. 4); the same frozen rules are applied to *every* model, including baselines that never saw them.

Algorithm 4 Rule calibration, rule-consistent training and violation rate.

Require: training evidence $\{Z_i\}$, labels $\{y_i\}$, embeddings $\{\varphi_i\}$, σ_y , λ

- 1: **Calibrate rules:** initialise θ ; minimise $\mathcal{L}_{\text{fit}}(\theta)$ (Eq. 13) with Adam; freeze θ
 - 2: **for** epoch = 1, ..., 60 **do**
 - 3: **for** mini-batch B **do**
 - 4: $s \leftarrow g_\psi(\varphi, Z)$; $\hat{F} \leftarrow \sigma(s)$
 - 5: $\mathcal{L} \leftarrow \mathcal{L}_{\text{ord}}(\hat{F}, F^*(y; \sigma_y)) + \lambda \cdot \frac{1}{4} \sum_k \text{ReLU}(\hat{F}_k - R_k(Z; \theta))^2$
 - 6: update ψ (AdamW, cosine schedule)
 - 7: **end for**
 - 8: every 3 epochs: compute quadratic-weighted kappa on validation half A; keep best ψ
 - 9: **end for**
 - 10: **Violation rate** on test set: $\frac{1}{N} \sum_i \mathbb{I}[\exists k \leq \hat{y}_i : R_k(Z_i; \theta) < 0.5]$
-

4.9. Calibration and split-conformal referral

Temperature scaling. On validation half A we fit a single temperature $T^* \in [0.5, 3]$ that minimises the negative log-likelihood of the class probabilities obtained from the logits s_k/T (Guo et al., 2017); this changes confidences but not predicted grades. Expected calibration error (ECE) is computed with 15 equal-width confidence bins.

Split-conformal sets. Let $\hat{p}(x) \in \Delta^4$ denote the calibrated class probabilities. On validation half B (calibration set of size n , disjoint from model selection

and temperature fitting), we define the nonconformity score $\varsigma_i = 1 - \hat{p}_{y_i}(x_i)$ and the threshold

$$\hat{q}_\alpha = \varsigma_{(\lceil (n+1)(1-\alpha) \rceil)}, \quad \mathcal{C}_\alpha(x) = \{j : 1 - \hat{p}_j(x) \leq \hat{q}_\alpha\}, \quad (15)$$

where $\varsigma_{(m)}$ is the m -th smallest score. If calibration and test data are exchangeable, $\Pr[y \in \mathcal{C}_\alpha(x)] \geq 1 - \alpha$ marginally (Angelopoulos and Bates, 2023; Vovk et al., 2005). Sets of size one are confident decisions; larger sets signal ambiguity. We define a *referral rule*: refer the patient if $\mathcal{C}_\alpha(x)$ contains any grade ≥ 2 (referable DR); because the guarantee concerns the true grade, the probability that a truly referable patient is *not* referred is at most α *marginally over the population*, but it need not hold within each class. We therefore also report the empirical class-conditional referral sensitivity (Alg. 5).

Algorithm 5 Calibration and split-conformal referral.

Require: trained head, validation halves A and B, test image x , level α

- 1: $T^\star \leftarrow \arg \min_T \text{NLL}(\text{half A}; T)$
 - 2: for $i \in \text{half B}$: $\varsigma_i \leftarrow 1 - \hat{p}_{y_i}(x_i; T^\star)$
 - 3: $\hat{q}_\alpha \leftarrow$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest ς_i
 - 4: $\mathcal{C}_\alpha(x) \leftarrow \{j : 1 - \hat{p}_j(x; T^\star) \leq \hat{q}_\alpha\}$
 - 5: **refer** iff $\max \mathcal{C}_\alpha(x) \geq 2$
-

4.10. Image-quality gate

Under the six-class protocol, ungradable images (grade 5 in DDR) are detected by a separate binary classifier with the same MLP architecture as the baseline, trained on all training images with the target $\mathbb{K}[y = 5]$. Its decision threshold is chosen on validation data to maximise the macro-F1 of the six-class prediction; an image flagged as ungradable receives class 5, otherwise the ordinal grade is output. The gate is deliberately kept independent of the ordinal head, so that the five-class results are not affected by it.

4.11. Design rationale

Several design choices deserve a justification, both to make clear what they are meant to achieve and to allow readers to disagree with them.

Why evidence from a segmenter rather than an end-to-end network? In DDR, only 5.5% of images have pixel-level lesion labels, but these labels are the most informative annotation available: they say *where* the clinically relevant structures are. A segmenter trained on them is an explicit way of transferring that supervision to the 95% of images that only carry a grade. An end-to-end network with the same information would have to discover the lesions from the grade, which is a harder, weaker signal. The price is that errors of the segmenter are propagated to the grade (Section 6.9).

Why soft counts, peaks and maxima, and why zones? Hard connected-component counts are sensitive to thresholds and are not differentiable; soft counts at three thresholds provide a multi-level summary that remains smooth, the peak count approximates the number of lesions rather than their area, and the maximum retains the evidence of the single most confident lesion. Zones are included because severity depends on how widely lesions are distributed (the 4–2–1 rule), which a single global count cannot express. We test the contribution of zones and of lesion classes in Section 6.8.

Why a transformer over tokens rather than an MLP over a vector? The 24 tokens are exchangeable up to the learned class and zone embeddings, so that attention can express interactions (for instance, haemorrhages in several quadrants combined with exudates) while remaining inspectable. The practical consequence is small in accuracy terms (Section 6.4), and we regard interpretability as the main reason for the choice.

Why necessary conditions, and why learn the thresholds? Clinical grading definitions are largely statements of the form “grade $\geq k$ requires a certain pattern”, i.e. necessary conditions for a high grade, and not sufficient ones, because a grader may call a lower grade for a reason that the lesion statistics do not capture. Encoding them as upper bounds on $\hat{F}_k$ therefore penalises only over-grading relative to the evidence. Thresholds are learned from labels, because the evidence statistics are those of a particular, imperfect segmenter and no clinical threshold is defined in terms of them.

Why split-conformal prediction on top of temperature scaling? Temperature scaling fixes confidence but offers no statement about error rates on new data. Split-conformal prediction offers a finite-sample marginal guarantee with a single hyper-parameter and no retraining, which suits a setting in which the base model is cheap to retrain but the calibration data are scarce. The price is the exchangeability assumption, whose failure we document in Section 6.6.

Why a frozen embedding? This is dictated by compute, but it also isolates the contribution of lesion evidence: if the image representation is held fixed, any gain is attributable to how lesion information is used. The corresponding drawback, that a fine-tuned backbone could capture some of the same information, is discussed in Sections 6.10 and 7.5.

4.12. Complexity

The segmentation network (11.8 M parameters) is the dominant cost: on four CPU cores with bfloat16 it processes about 8 images per second at 512×512 (Section 5). Evidence extraction (Alg. 3) costs $\mathcal{O}(4 \cdot 6 \cdot hw)$ per image and is negligible. The grading head has 0.14 M parameters, and each ordinal prediction needs 26 tokens, i.e. $\mathcal{O}(26^2 d)$ attention operations. Conformal calibration needs a sort of n scores. The method thus adds under 1% of parameters and well under 1% of the compute of the segmentation network on top of a backbone embedding.

5. Experimental setup

5.1. Compute environment and its consequences

All experiments were run on a single cloud virtual machine with four CPU cores of an Intel Xeon (2.1 GHz, AVX-512 and AMX bfloat16 instructions), 15 GB of memory and no GPU. PyTorch (Paszke et al., 2019) with bfloat16 autocast and channels-last tensors gave a 3–4 $\times$ speed-up over float32 on this hardware, which made the following design feasible: the segmentation network trains in about 23 minutes (30 epochs) and processes about eight 512×512 images per second at inference. Image embeddings come from a *frozen* ImageNet-pretrained backbone, so that the grading stage trains in seconds and can be repeated over many seeds. This design imposes limitations that readers should keep in mind: (i) the segmentation network is trained at 512×512 rather than at native resolution (about 1950×1950), which shrinks microaneurysms to one to three pixels; (ii) it is trained on the 383 lesion-training images only; (iii) the image backbone is not fine-tuned on DR; (iv) segmentation results come from a single training run per configuration (seed 0), while grading-stage results average five seeds; and (v) heavy detectors (Faster R-CNN, DETR) and large transformer backbones could not be trained within the budget, so detection is evaluated through segmentation-derived proposals against a classical image-processing baseline and across segmentation losses.

5.2. Implementation details

Segmentation. Architecture, losses, sampling and optimisation are as described in Section 4.3. Variants trained: (a) $\mathcal{L}_{\text{bce}}$, (b) $\mathcal{L}_{\text{bce+dice}}$, (c) $\mathcal{L}_{\text{seg}}$ (focal Tversky) on raw colour input, and (d) $\mathcal{L}_{\text{seg}}$ on Gaussian-filtered input (“GF”; Section 3.3). All use the same seed, crop sampler and schedule. Thresholds for Dice/IoU are chosen on lesion-validation data per class.

Image embedding. ConvNeXt-T (ImageNet-1k weights `convnext_tiny.fb_in1k` from `timm`) is applied to 256×256 Gaussian-filtered images; global averages of stage 3 (384 channels) and stage 4 (768 channels) are concatenated to a 1152-dimensional vector and standardised using training-set statistics. We chose this backbone and resolution a priori for throughput reasons; no alternative was evaluated on the test set.

Grading heads. All heads are trained with AdamW (weight decay 0.01), cosine schedule over 60 epochs, mini-batches of 128, with learning rate 10^{-3} for MLPs and 5×10^{-4} for the token head; the epoch with the highest quadratic-weighted kappa on validation half A (checked every three epochs) is retained. Validation is split into two halves with a fixed random permutation: half A (1251 images) for model selection and temperature, half B (1252 images) for conformal calibration. Each configuration is trained with five seeds (0–4); the data split is the same for all.

5.3. Research questions

The experiments are organised around six questions, each tied to one of the gaps of Section 1; we state them in advance so that results are read against explicit expectations rather than selected afterwards.

RQ1 (G3, segmentation). Does a class-weighted overlap objective with lesion-centred sampling improve small-lesion segmentation relative to plain cross-entropy and to a classical filter, and does Gaussian-filtered input matter?

RQ2 (G3, detection). How far can segmentation-derived proposals go as lesion detectors at relaxed and strict overlap, and what limits them?

RQ3 (G1, lesion–grade coupling). Does zone-wise lesion evidence add grading information beyond a frozen image embedding, and does the token head help beyond naive fusion?

RQ4 (G2, ordinal noise; G1, consistency). Do Gaussian-smoothed ordinal targets and the clinical-rule consistency loss improve agreement, calibration, or the rule-violation rate?

RQ5 (G4, referral). Do conformal sets reach their nominal coverage, and does the lesion-grounded model give better referral operating points than baselines?

RQ6 (robustness). Do the conclusions hold under a change of population, camera and resolution (zero-shot external test), and against an end-to-end fine-tuned reference?

5.4. Compared methods

Table 4 lists the grading models. B1–B4 share the same embedding and (where applicable) the same lesion evidence as the proposed models, so that every difference is attributable to *how* lesion information is used.

For segmentation, the four U-Net variants of Section 4.3 are compared. For detection we compare segmentation-derived proposals from each variant and a classical baseline: top-hat filtering of the green channel followed by threshold and connected components (dark top-hat for MA and HE, bright top-hat for EX and SE), with the threshold tuned per class on lesion-validation data. We do not compare against published numbers of other systems because their splits, resolutions and training budgets differ.

5.5. Evaluation metrics

Grading. The primary metric is the quadratic-weighted Cohen’s kappa (QWK) (Cohen, 1968), the standard agreement measure for ordinal DR grades. We also report accuracy, macro-averaged F1, the area under the ROC curve (AUC) for referable DR ($y \geq 2$) computed from $P(y \geq 2)$, expected calibration error (ECE, 15 bins), and the rule-violation rate (Section 4.8). Conformal prediction is evaluated at $\alpha \in \{0.05, 0.1, 0.2\}$ by empirical coverage, mean set size, singleton fraction and referral sensitivity/specificity. All grading metrics are computed on the leak-controlled test set (Section 3) unless stated.

Segmentation. The area under the precision–recall curve (AUPR) of the pixel probabilities, computed exactly after quantising probabilities to 512 levels, is the primary metric because it is threshold-free and appropriate for extreme class imbalance (Saito and Rehmsmeier, 2015). We additionally report Dice

Table 4: Grading models compared. All use the same frozen ConvNeXt-T embedding $\varphi(x)$; “evidence” means the tensor Z derived from the segmentation maps of the same segmentation network.

Model	Description	Evidence	Ordinal	Soft	Rule
B1	MLP on $\varphi(x)$, 5-way cross-entropy	–	–	–	–
B2	MLP on $\varphi(x)$, cumulative-link loss (Eq. 8, $\sigma_y = 0$)	–	✓	–	–
B3	MLP on the flattened evidence Z only	✓	✓	–	–
B4	MLP on concatenation $[\varphi(x), Z]$ (naive fusion)	✓	✓	–	–
T0	Evidence-token transformer, $\sigma_y = 0$, $\lambda = 0$	✓	✓	–	–
T1	T0 with soft targets ($\sigma_y = 0.5$)	✓	✓	✓	–
T2	T0 with rule loss ($\lambda = 1$)	✓	✓	–	✓
LesionRule	transformer, $\sigma_y = 0.5$, $\lambda = 1$	✓	✓	✓	✓
A1	LesionRule without evidence tokens in the head (image token only; rule loss still reads Z)	(rule only)	✓	✓	✓

(Dice, 1945) and intersection-over-union (IoU) at the per-class validation-optimal threshold, accumulated over all test pixels.

Detection. Average precision at IoU 0.5 and 0.3 (all-point interpolation (Padilla et al., 2021)) and lesion-level FROC sensitivity at 1, 2, 4, 8 and 16 false positives per image, where a prediction is a hit if its centre lies in a ground-truth box. The relaxed IoU of 0.3 is reported because, for a box a few pixels wide, one pixel of displacement lowers IoU dramatically.

5.6. Statistical analysis

Grading results are reported as mean $\pm$ standard deviation over five seeds. For the main comparisons we additionally report 95% confidence intervals obtained by a non-parametric bootstrap of the test images (Efron and Tibshirani, 1993) (2000 resamples) applied to the seed-averaged class probabilities,

and paired bootstrap differences in QWK between models, which respect the pairing of test images. Differences are deemed significant when the 95% interval of the paired difference excludes zero; no multiplicity correction is applied to the exploratory ablation table, and we say so where relevant. Segmentation metrics are reported from a single run per configuration; we therefore do not make significance claims among segmentation variants and we compare them descriptively.

5.7. Reproducibility

All scripts (data extraction, training, evaluation, figure and table generation) and intermediate artefacts (cached lesion maps, embeddings, per-seed probabilities) are released with the paper. Every number in the tables is generated by scripts from stored result files; there is no manual transcription. Random seeds are fixed. The code reads the DDR archive directly from its split zip volumes, so no re-packaging is needed.

6. Results

We report segmentation and detection first (Sections 6.1 and 6.2), since they determine the quality of the lesion evidence, then the grading results (Sections 6.3–6.7), the end-to-end reference (Section 6.10) and computational cost (Section 6.11). Unless stated otherwise, segmentation and detection use the 225-image DDR lesion-test split, and grading uses the leak-controlled test set of 3633 gradable images.

6.1. Lesion segmentation

Table 5 compares the five segmentation systems on the 225 DDR test images, and Fig. 6 shows examples. Training curves are in Fig. 7.

Learned versus classical. The classical top-hat baseline is nearly useless at this resolution (mean AUPR 0.023), which is expected for hand-crafted detectors on fundus images with illumination changes, vessels and the optic disc. All four U-Net variants are an order of magnitude better (mean AUPR 0.30–0.38).

Loss function. Plain cross-entropy fails on the smallest and rarest lesions: its AUPR for microaneurysms is only 0.025 (Dice 0.054), compared with 0.156–0.172 for the three class-weighted overlap objectives, and its mean AUPR is 0.303 versus 0.361–0.382. For hard exudates, which are frequent and large, cross-entropy is competitive (AUPR 0.588, the highest in the table). The three overlap-based objectives are close to each other. BCE+Dice has the highest mean AUPR (0.382) and the focal Tversky objective the highest mean Dice at the validation-optimal threshold (0.432 raw input, 0.440 with Gaussian-filtered input, versus 0.429 for BCE+Dice). Since every row is a single run, we do not claim that any of the three is better than the others; the robust conclusion is that *a class-weighted overlap term is necessary for micro-aneurysms and soft exudates*, whereas the specific form (Dice or focal Tversky) is of second order in our experiment. We stress this because the focal Tversky loss was our default choice and it did not prove superior.

Gaussian-filtered input. Feeding the Gaussian-filtered image to the segmenter (“GF”) gives practically the same performance as the raw image (mean AUPR 0.365 vs. 0.361; mean Dice 0.440 vs. 0.432), with small per-class differences of both signs (microaneurysm AUPR 0.172 vs. 0.156; haemorrhage 0.436 vs. 0.470). The GF variant was used for all downstream grading experiments because it matches the preprocessing of the external Kaggle set.

Class difficulty. The ranking of classes is stable across variants: exudates (AUPR 0.55–0.59) and haemorrhages (0.40–0.47) are segmented best, soft exudates are intermediate and variable (0.20–0.32), and microaneurysms are the hardest (at most 0.17 AUPR and 0.29 Dice). Microaneurysms are about 10 native pixels wide at the median and about 4–5 pixels at the 10th percentile (Section 3.1); after resizing to 512×512 most of them are one to three pixels wide, so part of the difficulty is a consequence of our compute-driven resolution choice. Examples in Fig. 6 show that the models capture the large-scale layout of exudates and haemorrhages well and typically over-segment (merge neighbouring lesions) or miss isolated small lesions. Fig. 8 shows precision–recall curves per class.

Validation versus test. Validation AUPR of the focal Tversky models (mean 0.53–0.54 at the selected checkpoints) is much higher than their test AUPR (0.36–0.37). As with the grading partitions (Section 6.6), the official partitions of the lesion subset therefore differ in difficulty and composition (for example, the lesion-test split has 6457 haemorrhage boxes in 225 images,

against 4739 in the 383 training images), and results tuned on validation data are optimistic.

Table 5: Lesion segmentation on the DDR test split (225 images, 512×512). AUPR is threshold-free; Dice and IoU are computed at the per-class threshold that maximises validation Dice and accumulated over all test pixels. GF = Gaussian-filtered input. Single training run per row (seed 0); best value per column in bold.

Method	AUPR					Dice					mIoU
	MA	HE	EX	SE	Mean	MA	HE	EX	SE	Mean	
Top-hat + threshold (classical)	0.001	0.025	0.058	0.006	0.023	0.003	0.063	0.097	0.017	0.045	0.024
U-Net, BCE	0.025	0.402	0.588	0.196	0.303	0.054	0.408	0.570	0.241	0.318	0.205
U-Net, BCE+Dice	0.160	0.469	0.582	0.317	0.382	0.268	0.498	0.591	0.358	0.429	0.281
U-Net, focal Tversky (default)	0.156	0.470	0.551	0.267	0.361	0.273	0.501	0.586	0.370	0.432	0.283
U-Net, focal Tversky, GF input	0.172	0.436	0.562	0.290	0.365	0.292	0.467	0.597	0.403	0.440	0.288

6.2. Lesion detection

Tables 6 and 7 report the detections obtained from the segmentation maps (Alg. 2), and Fig. 9 shows examples.

Overall level. Detection through connected-component proposals is weak at the standard IoU of 0.5, and moderate at relaxed overlap. The mean AP at IoU 0.3 is 0.168 for the GF variant (0.161 raw; 0.168 BCE+Dice; 0.109 BCE; 0.019 for the classical baseline). At IoU 0.5 the AP of micro-aneurysms is below 0.01: a box that is one or two pixels off in each direction already drops below the threshold, which is a property of the metric on tiny objects more than of the detector, and the AP of haemorrhages and exudates is 0.06–0.08. Soft exudates reach $AP_{0.5}$ of 0.155–0.168, in line with their larger size but with only 214 test boxes the estimate is noisy.

FROC. At 8 false positives per image, the sensitivities of the best models are about 0.29–0.32 for microaneurysms, 0.31–0.32 for haemorrhages, 0.31–0.34 for exudates and 0.43–0.46 for soft exudates (Table 7); they saturate at 16 false positives per image (0.32–0.41), which reflects that lesions missed by the segmentation (and therefore never proposed) bound the recall. The ordering of segmentation variants is the same as in segmentation: the plain cross-entropy model is clearly worse ($mAP_{0.3}$ 0.109) and the other three are close (0.161–0.168).

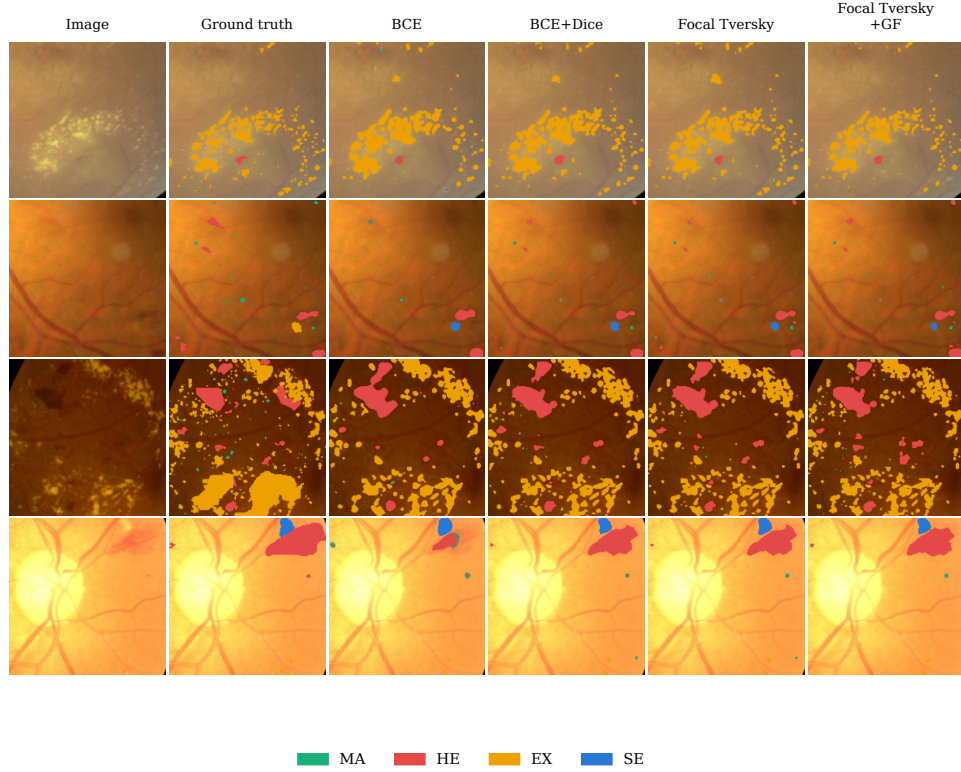


Figure 6: Qualitative segmentation results on four DDR test images (crops of 192×192 pixels containing many lesions), with predictions thresholded at each model’s validation-optimal threshold. Columns: input, ground truth, and the four U-Net variants. Colours: MA green, HE red, EX yellow, SE blue.

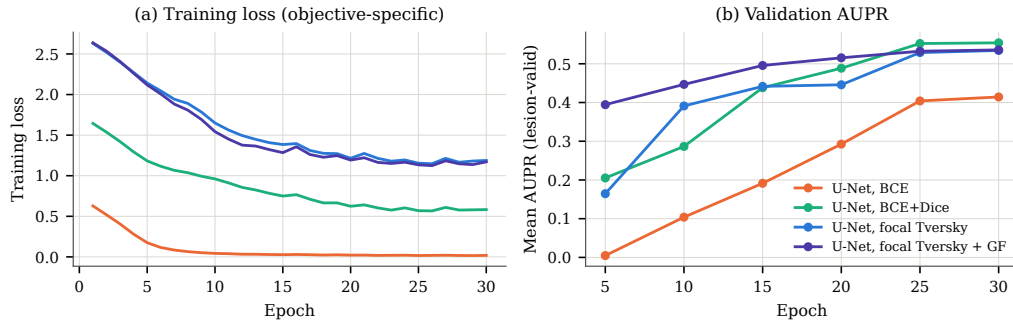


Figure 7: Segmentation training. (a) Training loss (objectives differ in scale and are not comparable across methods). (b) Mean AUPR over the four lesion classes on the lesion-validation split at the evaluation epochs.

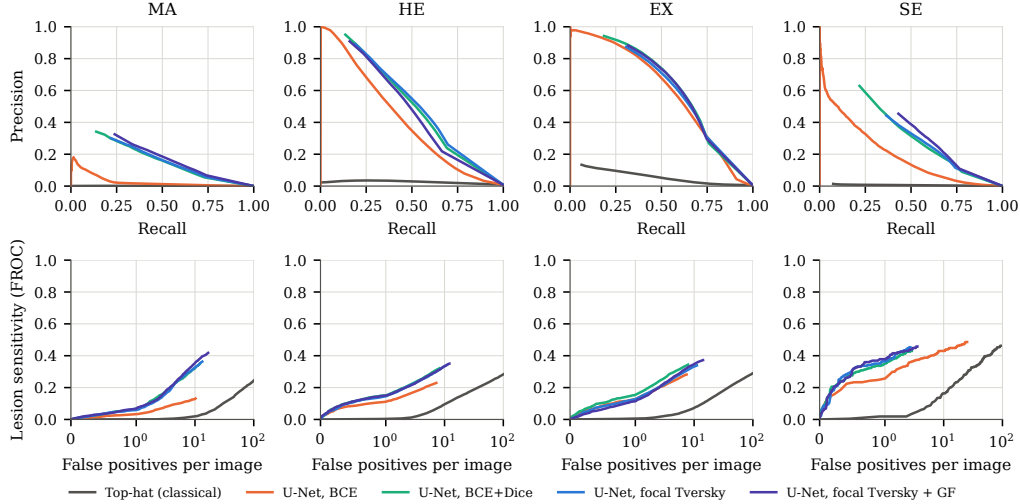


Figure 8: Per-class precision–recall curves of pixel probabilities (top) and lesion-level FROC curves of segmentation-derived detections (bottom; false positives per image on a symmetric-log axis), DDR test split.

Why detection is hard here. Several factors explain the low numbers: (i) very high lesion density (the test images have 76 boxes on average), so adjacent lesions merge into one connected component; (ii) boxes of the order of 10 pixels or less; (iii) annotation boxes derived from pixel masks, which are not independent of segmentation errors; and (iv) the use of a seed threshold selected on validation $AP_{0.3}$ (the selected seeds differ widely between classes, 0.05–0.71, indicating that the probability maps are not calibrated across classes). We did not train a dedicated detector (Section 5), so we cannot say how this approach compares with Faster R-CNN or FCOS; the value of the segmentation-derived route is that it requires no additional training and shares weights with the other tasks.

6.3. What the lesion maps look like, and how lesion evidence relates to grade

Fig. 10 shows the lesion maps predicted for one random test image per grade. Maps are almost empty for grade 0, contain scattered microaneurysm and exudate responses for grades 1–2, and become dense and multi-class for the higher grades. The grade-4 example also illustrates a failure mode: a large pre-retinal structure in this image attracts haemorrhage and exudate responses over a wide area, which is not a “lesion count” in the clinical sense

Table 6: Lesion detection on the DDR test split from segmentation-derived proposals (Alg. 2). $AP_{0.5}$ / $AP_{0.3}$ are average precision at IoU 0.5 / 0.3; the last column is the mean of $AP_{0.3}$ over the four classes. The ground truth contains 2041 MA, 6457 HE, 8320 EX and 214 SE boxes.

Proposals from	MA		HE		EX		SE		mAP _{0.3}
	AP _{0.5}	AP _{0.3}	AP _{0.5}	AP _{0.3}	AP _{0.5}	AP _{0.3}	AP _{0.5}	AP _{0.3}	
Top-hat + threshold (classical)	0.000	0.005	0.010	0.029	0.015	0.038	0.003	0.004	0.019
U-Net, BCE	0.001	0.012	0.042	0.114	0.067	0.167	0.122	0.142	0.109
U-Net, BCE+Dice	0.008	0.081	0.078	0.180	0.084	0.215	0.168	0.197	0.168
U-Net, focal Tversky (default)	0.008	0.077	0.080	0.184	0.062	0.193	0.155	0.188	0.161
U-Net, focal Tversky, GF input	0.008	0.089	0.079	0.187	0.071	0.207	0.155	0.188	0.168

Table 7: Lesion-level FROC sensitivity of segmentation-derived detections (a prediction is a hit if its centre lies inside a ground-truth box) at 1, 2, 4, 8 and 16 false positives per image (FP/img), DDR test split.

Proposals from	Class	1	2	4	8	16
Top-hat + threshold (classical)	MA	0.000	0.001	0.004	0.014	0.040
	HE	0.003	0.007	0.028	0.076	0.135
	EX	0.004	0.011	0.027	0.058	0.113
	SE	0.019	0.019	0.056	0.136	0.234
U-Net, BCE	MA	0.032	0.056	0.090	0.119	0.131
	HE	0.111	0.147	0.188	0.229	0.229
	EX	0.128	0.178	0.233	0.282	0.282
	SE	0.257	0.332	0.379	0.421	0.449
U-Net, focal Tversky, GF input	MA	0.067	0.115	0.209	0.319	0.412
	HE	0.150	0.194	0.250	0.309	0.351
	EX	0.115	0.178	0.254	0.330	0.373
	SE	0.379	0.425	0.458	0.458	0.458

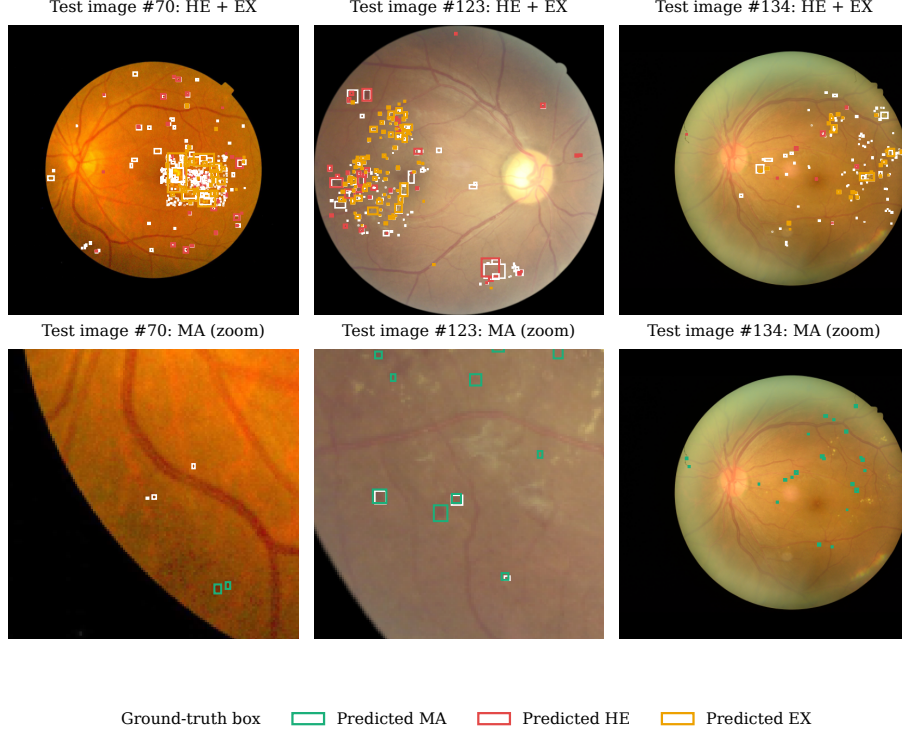


Figure 9: Segmentation-derived detections (GF model) on three DDR test images: ground-truth boxes (white) and predicted boxes (HE red, EX yellow, MA green; predictions with low score are hidden for legibility). Top: haemorrhages and exudates; bottom: microaneurysms in a magnified region.

(the aggregated statistics would overstate the burden). Because the maps are summarised into the evidence tensor Z , we can also look at the statistics directly. Fig. 11 shows, for each lesion class and zone, the mean $\log(1+n^{(0.5)})$ over the test images of each true grade. The soft counts of haemorrhages, hard exudates and soft exudates are clearly higher for grades 2–4 than for grades 0–1 in every zone, with the largest differences in the whole-field and quadrant statistics, and are not strictly monotone between the highest grades; the central-disc statistic is smaller than the others. Microaneurysm evidence is the least discriminative class (its mean varies little with grade), consistent with its weak segmentation quality (Section 6.1). The statistics therefore carry the signal that the rules (Eqs. 9–12) rely on, but they are noisy at individual-image level.

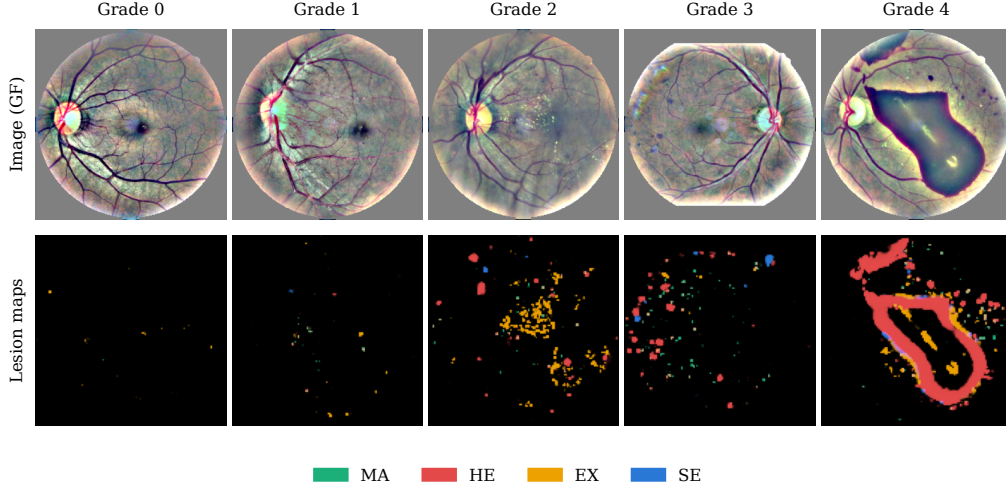


Figure 10: Gaussian-filtered inputs (top) and predicted lesion probability maps (bottom, 128×128 after max-pooling, colour = class as in Fig. 2) for one randomly chosen test image of each grade. The grade-4 image shows a large non-lesion structure producing wide responses.

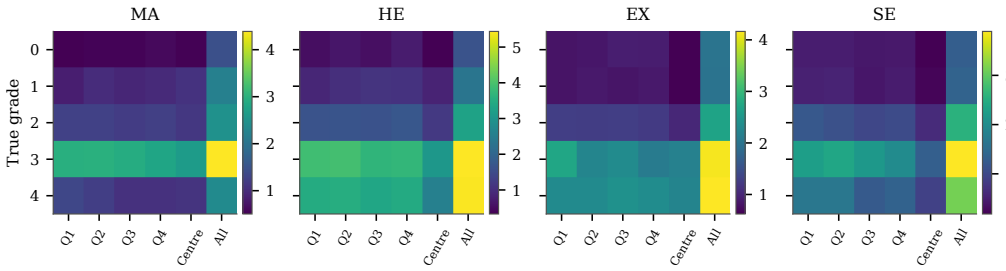


Figure 11: Mean zone-wise lesion evidence $\log(1 + n_{c,z}^{(0.5)})$ by true grade (rows) and zone (columns: four quadrants Q1–Q4, central disc, whole field) on the leak-controlled test set, one panel per lesion class.

6.4. Grading performance

Table 9 and Fig. 12 report the five-class grading results. We discuss four findings in turn.

(1) *Lesion evidence improves grading substantially.* The two models that use only the frozen image embedding (B1, B2) reach a QWK of 0.644 and 0.638. A model that sees *only* the 120-dimensional evidence tensor (B3) reaches the same agreement (0.641) with 0.016 M instead of 0.30 M parameters, which says that the lesion maps contain as much grading information as a generic 1152-dimensional ImageNet embedding. Combining the two is clearly better than either alone: the naive concatenation B4 reaches 0.722, and LesionRule reaches 0.734 (± 0.009 over five seeds; 95% bootstrap interval [0.714, 0.754]). The paired bootstrap difference between LesionRule and B1/B2/B3 is +0.087/+0.095/+0.092 with intervals that exclude zero by a wide margin (Table 9). The AUC for referable DR rises from 0.865–0.867 (B1, B2) to 0.923. Ablation A1, which keeps everything except the evidence tokens in the head, falls back to 0.632, i.e. to the level of B2 and *not* to that of the full model, which shows that the gain comes from the evidence *as input* to the head, not from the soft labels or the rule loss alone.

(2) *Evidence tokens versus naive concatenation: a small gain.* The token-based head is better than concatenating the same inputs, but only slightly: the paired difference to B4 is +0.013 in QWK (95% interval [0.001, 0.025]) and to the token head without soft labels or rule loss (T0) +0.008 ([0.000, 0.017]). With five seeds and a single backbone, we regard this as weak evidence that the token structure helps beyond the information content of the evidence itself; its principal benefit, in our view, is interpretability (attention to (c, z) tokens) rather than accuracy.

(3) *Soft ordinal labels mainly improve calibration, not agreement.* Adding Gaussian-smoothed targets (T1) changes QWK by +0.010 relative to T0 (seed-level SDs are 0.014), i.e. not reliably, but it reduces the expected calibration error from 0.095 to 0.058 (and to 0.057 for the full model, versus 0.093 for B1 and 0.120 for B2). This is consistent with the intended role of the soft target: it prevents the cumulative logits from saturating when adjacent grades are confused.

(4) *The rule loss gives no measurable accuracy gain on DDR.* T2 (rule loss, no soft labels) scores 0.729 versus 0.722 for T0, and the full model is within 0.001 of T1 (paired difference +0.001, interval $[-0.008, 0.010]$). The reason is visible in Section 6.5: grade predictions of *all* models are already consistent with the fitted rules in more than 94% of cases, so there is little left to correct.

Per-class behaviour. Fig. 13a shows where the errors are. Grade 0 is recognised with recall 0.98, and moderate (0.58) and proliferative DR (0.58) with precision around 0.75–0.78. Mild DR (grade 1) and severe NPDR (grade 3) are the weak classes (recall 0.04 and 0.07; F1 0.06 and 0.12): grade 1 is mostly predicted as grade 0 or 2, and grade 3 as grade 2 or 4. These are the two rarest classes (1.7% and 4.6% of training images for grades 3 and 1, respectively) and are defined by fine distinctions (a few microaneurysms; the 4–2–1 rule), so a low macro-F1 (0.46) is unsurprising, but it is a clear limitation. Overall, 82.5% of the predictions are within one grade of the label, and the dominant error mode is the confusion of grade 2 with grade 0 (37% of grade-2 images are called normal). On the six-class protocol (Table 10) accuracy is 0.734 for LesionRule versus 0.692 (B1); the quality gate separates ungradable from gradable images with AUC 0.995 for every model.

Table 8: Per-class results of LesionRule on the leak-controlled test set (seed-averaged probabilities, 3633 images).

	G0	G1	G2	G3	G4
Support	1880	163	1265	61	264
Precision	0.77	0.12	0.75	0.67	0.78
Recall	0.98	0.04	0.58	0.07	0.58
F1	0.86	0.06	0.66	0.12	0.66

6.5. Rule consistency

Under the frozen rule scores, the fraction of test predictions that exceed what the lesion evidence licenses is 3.9–5.2% for all models (Table 9; Fig. 12, right). The full model has 4.5% and the model without lesion evidence in the head, B2, has 4.4%; the token model without rule loss (T0) has the highest rate (5.2%) and the rule-loss variants (T2, full) are 0.7–0.8 percentage points lower, a small shift that is of the same order as the seed-to-seed spread (SD up to 0.8 points). We therefore cannot claim that the rule-consistency loss reduces contradictions in a meaningful way on this data. Two explanations are

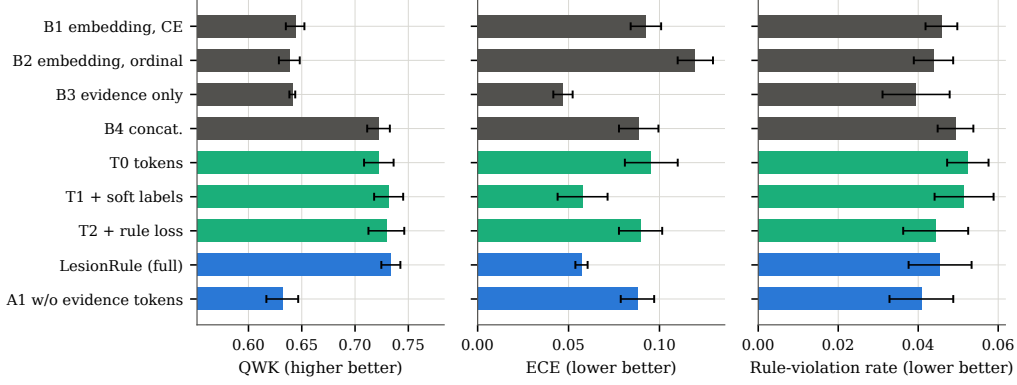


Figure 12: Grading models of Table 4 (mean $\pm$ SD over five seeds): quadratic-weighted kappa, expected calibration error and rule-violation rate. Grey: baselines using the embedding and/or evidence in simple form; green: evidence-token models; blue: full LesionRule and the ablation without evidence tokens (A1).

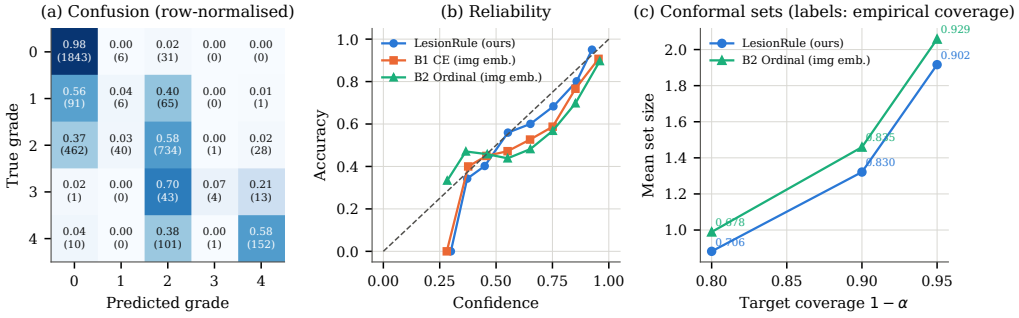


Figure 13: (a) Row-normalised confusion matrix of LesionRule with counts. (b) Reliability diagram (10 bins) for LesionRule and two baselines. (c) Mean conformal set size against target coverage (calibration on the validation partition); numbers give the empirical coverage on the test set, which is below the target (Section 6.6).

Table 9: Five-class DR grading on the leak-controlled DDR test set (3633 gradable images), mean $\pm$ SD over 5 seeds. QWK 95% CI: bootstrap over test images of the seed-averaged probabilities. Δ QWK: paired bootstrap difference (LesionRule minus model), with 95% interval; intervals excluding 0 are marked *. Rule viol.: fraction of test images whose predicted grade is not licensed by the frozen rule scores (Section 4.8).

Model	QWK	95% CI	Acc.	F1	AUC _{≥ 2}	ECE	Rule viol.	Δ QWK
B1: embedding, CE	0.644 $\pm$ 0.009	[0.623, 0.669]	0.686 $\pm$ 0.006	0.439 $\pm$ 0.011	0.865 $\pm$ 0.003	0.093 $\pm$ 0.008	0.046 $\pm$ 0.004	+0.087 [+0.071, +0.103]*
B2: embedding, ordinal	0.638 $\pm$ 0.010	[0.616, 0.662]	0.683 $\pm$ 0.005	0.428 $\pm$ 0.008	0.867 $\pm$ 0.005	0.120 $\pm$ 0.010	0.044 $\pm$ 0.005	+0.095 [+0.079, +0.111]*
B3: evidence only	0.641 $\pm$ 0.003	[0.619, 0.665]	0.697 $\pm$ 0.005	0.449 $\pm$ 0.007	0.877 $\pm$ 0.002	0.047 $\pm$ 0.005	0.039 $\pm$ 0.008	+0.092 [+0.072, +0.112]*
B4: embedding + evidence (concat.)	0.722 $\pm$ 0.011	[0.702, 0.741]	0.738 $\pm$ 0.007	0.486 $\pm$ 0.007	0.920 $\pm$ 0.003	0.089 $\pm$ 0.011	0.049 $\pm$ 0.004	+0.013 [+0.001, +0.025]*
T0: evidence tokens	0.722 $\pm$ 0.014	[0.706, 0.747]	0.748 $\pm$ 0.010	0.454 $\pm$ 0.017	0.921 $\pm$ 0.004	0.095 $\pm$ 0.015	0.052 $\pm$ 0.005	+0.008 [+0.000, +0.017]*
T1: T0 + soft labels	0.732 $\pm$ 0.014	[0.714, 0.753]	0.745 $\pm$ 0.007	0.472 $\pm$ 0.022	0.923 $\pm$ 0.002	0.058 $\pm$ 0.014	0.051 $\pm$ 0.007	+0.001 [-0.008, +0.010]
T2: T0 + rule loss	0.729 $\pm$ 0.017	[0.714, 0.753]	0.745 $\pm$ 0.008	0.457 $\pm$ 0.025	0.921 $\pm$ 0.005	0.090 $\pm$ 0.012	0.044 $\pm$ 0.008	+0.001 [-0.007, +0.009]
LesionRule (full)	0.734 $\pm$ 0.009	[0.714, 0.754]	0.745 $\pm$ 0.011	0.464 $\pm$ 0.018	0.923 $\pm$ 0.004	0.057 $\pm$ 0.003	0.045 $\pm$ 0.008	
A1: LesionRule w/o evidence tokens	0.632 $\pm$ 0.015	[0.618, 0.664]	0.682 $\pm$ 0.003	0.408 $\pm$ 0.017	0.851 $\pm$ 0.006	0.088 $\pm$ 0.009	0.041 $\pm$ 0.008	+0.093 [+0.077, +0.109]*

Table 10: Six-class protocol (ungradable images included) on the leak-controlled test set: a separate quality gate decides “ungradable; otherwise the ordinal grade is output. Mean $\pm$ SD over 5 seeds.

Model	Accuracy	Macro-F1	Gate AUC
B1: embedding, CE	0.692 $\pm$ 0.005	0.488 $\pm$ 0.012	0.995 $\pm$ 0.000
B2: embedding, ordinal	0.676 $\pm$ 0.004	0.495 $\pm$ 0.007	0.995 $\pm$ 0.000
B3: evidence only	0.678 $\pm$ 0.006	0.505 $\pm$ 0.004	0.995 $\pm$ 0.000
B4: embedding + evidence (concat.)	0.726 $\pm$ 0.006	0.543 $\pm$ 0.006	0.995 $\pm$ 0.000
T0: evidence tokens	0.736 $\pm$ 0.007	0.530 $\pm$ 0.012	0.995 $\pm$ 0.000
T1: T0 + soft labels	0.735 $\pm$ 0.007	0.532 $\pm$ 0.011	0.995 $\pm$ 0.000
T2: T0 + rule loss	0.733 $\pm$ 0.008	0.538 $\pm$ 0.016	0.995 $\pm$ 0.000
LesionRule (full)	0.734 $\pm$ 0.010	0.533 $\pm$ 0.010	0.995 $\pm$ 0.000
A1: LesionRule w/o evidence tokens	0.669 $\pm$ 0.003	0.482 $\pm$ 0.011	0.995 $\pm$ 0.000

consistent with the numbers. First, the rules were calibrated on the training labels as necessary conditions (Eq. 13), so they are permissive by construction: the threshold of each rule settles where it is hardly ever violated by true labels, and because every model has learned this association from the same evidence, very few predictions fall outside it. Second, the weakest classes (mild and severe) are rarely predicted, which removes many opportunities to violate rules defined for $y \geq 1$ and $y \geq 3$. Stricter, anatomically defined rules (Section 7.5) or hand-specified thresholds would be needed to test the idea properly.

6.6. Calibration and conformal referral

Calibration. After temperature scaling on validation half A, the ECE of the full model is 0.057, versus 0.093 for the cross-entropy MLP and 0.120 for the ordinal MLP (Fig. 13b). The reliability curves lie close to the diagonal for confidence above 0.5, with the largest deviations in sparsely populated low-confidence bins.

Conformal sets calibrated on the DDR validation partition. Table 11 shows that the conformal sets calibrated on validation half B *do not* reach the nominal coverage on the test set: with $\alpha = 0.05, 0.1$ and 0.2 the empirical coverage of LesionRule is 0.902, 0.830 and 0.706 (targets 0.95, 0.90, 0.80), and the other models behave similarly. This is not a failure of the method but of its assumption: the DDR validation and test partitions are *not exchangeable*. Table 13 shows that the same model reaches an accuracy of 83.9% on validation half B but 75.4% on the test set, although the class proportions are practically identical (e.g. 51% versus 52% grade 0), with similar differences in QWK and log-loss. The nonconformity scores of the calibration set are therefore systematically smaller than those of the test set, and the quantile is too optimistic. We could not determine the origin of this shift (the dataset documentation does not describe how the partitions were formed; differences in hospital, camera or annotation batch are plausible).

Conformal sets under exchangeability. To verify that the construction itself works, Table 12 calibrates on a random half of the test set and evaluates on the other half (300 random splits). Coverage is then 0.950, 0.900 and 0.800 at the three nominal levels, with a standard deviation of about 0.01, as the theory (Appendix Appendix C) predicts. In this setting the referral rule (“refer if the set contains any grade ≥ 2 ”) at $\alpha = 0.1$ achieves a sensitivity of 0.845 and a specificity of 0.876 for the lesion-grounded model, compared with 0.882 and 0.631 for the embedding-only ordinal model B2 (and 0.871/0.813 for concatenation B4), at similar mean set sizes (1.90 versus 1.84 and 1.64): the lesion evidence makes the model’s *uncertainty* more informative, so that fewer healthy patients are referred at a similar sensitivity. For comparison, simply taking the arg-max grade gives a referable-DR sensitivity of 0.677 and a specificity of 0.953 (B2: 0.538 and 0.951), i.e. hard decisions miss a large fraction of referable cases at this operating point, which is the practical argument for set-valued outputs. We emphasise that the exchangeable check used test labels for calibration and therefore describes what would happen

with a calibration set drawn from the deployment distribution; it is not a test-set result in the usual sense.

Table 11: Split-conformal prediction sets (calibration: validation half B, 1252 images) evaluated on the leak-controlled test set. Coverage should be $\geq 1 - \alpha$. “Refer = set contains any grade ≥ 2 ; sensitivity/specificity are for referable DR ($y \geq 2$). Class-wise coverage shows that the guarantee is marginal.

Model	α	Coverage	Set size	Singleton	Refer sens.	Refer spec.	Cov. G0/G1/G2/G3/G4
LesionRule (full)	0.05	0.902	1.92	0.34	0.849	0.874	0.99/0.92/0.80/0.80/0.74
	0.1	0.830	1.32	0.75	0.767	0.922	0.99/0.44/0.70/0.43/0.65
	0.2	0.706	0.88	0.88	0.560	0.971	0.97/0.00/0.48/0.00/0.51
B1: embedding, CE	0.05	0.915	1.96	0.35	0.906	0.574	1.00/0.38/0.88/0.69/0.87
	0.1	0.829	1.43	0.63	0.771	0.798	1.00/0.10/0.71/0.46/0.75
	0.2	0.679	0.98	0.95	0.514	0.944	0.97/0.00/0.38/0.13/0.58
B2: embedding, ordinal	0.05	0.929	2.06	0.34	0.925	0.535	1.00/0.49/0.91/0.75/0.86
	0.1	0.835	1.46	0.62	0.782	0.784	1.00/0.10/0.72/0.61/0.73
	0.2	0.678	0.99	0.96	0.512	0.947	0.98/0.00/0.38/0.11/0.53
B4: embedding + evidence (concat.)	0.05	0.923	1.81	0.48	0.899	0.770	1.00/0.63/0.87/0.77/0.85
	0.1	0.838	1.30	0.74	0.795	0.900	1.00/0.17/0.73/0.52/0.73
	0.2	0.709	0.93	0.93	0.577	0.960	0.98/0.01/0.46/0.18/0.51

Table 12: Conformal coverage under exchangeability: the leak-controlled test set is split at random into a calibration half and an evaluation half (300 repetitions; mean $\pm$ SD). Compare with Table 11, where calibration used the DDR validation partition. “Refer = set contains any grade ≥ 2 .

Model	α	Coverage	Set size	Refer sens.	Refer spec.
LesionRule (full)	0.05	0.950 $\pm$ 0.008	2.75	0.913	0.770
	0.1	0.900 $\pm$ 0.011	1.90	0.845	0.876
	0.2	0.800 $\pm$ 0.014	1.17	0.733	0.936
B2: embedding, ordinal	0.05	0.950 $\pm$ 0.007	2.29	0.952	0.444
	0.1	0.900 $\pm$ 0.010	1.84	0.882	0.631
	0.2	0.799 $\pm$ 0.013	1.33	0.726	0.839
B4: embedding + evidence (concat.)	0.05	0.950 $\pm$ 0.007	2.10	0.934	0.676
	0.1	0.899 $\pm$ 0.010	1.64	0.871	0.813
	0.2	0.800 $\pm$ 0.013	1.17	0.745	0.920

6.7. Zero-shot external test

Table 14 gives the results of applying the DDR-trained heads to the Gaussian-filtered Kaggle set without adaptation. The transfer is poor for all models: QWK is between 0.10 and 0.23 and accuracy between 31% and

Table 13: Partition shift inside DDR: the same models evaluated on validation half B (the conformal calibration set, 1252 images) and on the leak-controlled test set (3633 images), seed-averaged probabilities. Class proportions are practically identical in the two sets.

Model	Acc. (val B)	Acc. (test)	QWK (val B)	QWK (test)
LesionRule (full)	0.839	0.754	0.857	0.735
B2: embedding, ordinal	0.802	0.685	0.810	0.640

50% depending on the seed, with a large seed-to-seed variance (e.g. the QWK of the full model over the five seeds is 0.38, 0.27, 0.05, 0.28 and -0.10). The AUC for referable DR is 0.64–0.73, only moderately above chance. Inspection of the predictions shows that the models collapse onto grades 0 and 2: for the full model (seed 0), 41% of the external images are predicted as grade 0, 59% as grade 2 and essentially none as grades 1, 3 or 4, although the external set has 27% grade-2 and 8% grade-4 images. Three factors plausibly contribute: (i) the external images are 224×224 , so microaneurysms and small haemorrhages are invisible and the lesion evidence is systematically underestimated; (ii) the segmentation network was trained on DDR images processed at full field of view and sees up-sampled, already-filtered inputs; (iii) the populations and cameras differ. The model with the rule loss and no soft labels (T2) has the highest mean QWK (0.227) and B2 reaches 0.215, but with five seeds and these standard deviations no ranking of models is supported. We report this negative result because it bounds the claims of the paper: *within DDR*, lesion evidence adds clearly to the embedding; *across datasets*, our pipeline does not generalise at this resolution, and the conformal guarantee could not be expected to hold either.

6.8. Which evidence matters? Zones and lesion classes

The design of the evidence bottleneck rests on two assumptions that the preceding results do not test: that resolving lesions by retinal zone helps, and that all four lesion classes contribute. Table 15 answers both questions by training the full LesionRule head with parts of the evidence tensor set to zero (including in the rule scores), using three seeds.

Zones add little.. Restricting the evidence to the whole-field zone gives a QWK of 0.736, compared with 0.745 for the reference with all six zones (a difference of -0.009), and removing the whole-field zone but keeping the four

Table 14: Zero-shot external test on the Gaussian-filtered APTOS-derived Kaggle set (3664 images, 224×224 , grades 0–4): models trained on DDR only, no adaptation (mean $\pm$ SD over 5 seeds). Lesion evidence is computed by the segmentation network applied to the up-sampled images.

Model	QWK	Acc.	F1	AUC $_{\geq 2}$	Rule viol.
B1: embedding, CE	0.175 $\pm$ 0.046	0.495 $\pm$ 0.030	0.219 $\pm$ 0.016	0.680 $\pm$ 0.037	0.018 $\pm$ 0.007
B2: embedding, ordinal	0.215 $\pm$ 0.087	0.455 $\pm$ 0.048	0.232 $\pm$ 0.034	0.694 $\pm$ 0.077	0.022 $\pm$ 0.006
B3: evidence only	0.103 $\pm$ 0.035	0.303 $\pm$ 0.011	0.141 $\pm$ 0.011	0.705 $\pm$ 0.001	0.015 $\pm$ 0.004
B4: embedding + evidence (concat.)	0.145 $\pm$ 0.027	0.348 $\pm$ 0.011	0.181 $\pm$ 0.020	0.726 $\pm$ 0.061	0.021 $\pm$ 0.005
T0: evidence tokens	0.121 $\pm$ 0.161	0.392 $\pm$ 0.094	0.188 $\pm$ 0.041	0.641 $\pm$ 0.125	0.020 $\pm$ 0.010
T1: T0 + soft labels	0.121 $\pm$ 0.189	0.375 $\pm$ 0.107	0.190 $\pm$ 0.054	0.642 $\pm$ 0.155	0.025 $\pm$ 0.011
T2: T0 + rule loss	0.227 $\pm$ 0.103	0.408 $\pm$ 0.069	0.206 $\pm$ 0.042	0.731 $\pm$ 0.089	0.021 $\pm$ 0.009
LesionRule (full)	0.176 $\pm$ 0.174	0.381 $\pm$ 0.079	0.199 $\pm$ 0.035	0.657 $\pm$ 0.147	0.022 $\pm$ 0.011
A1: LesionRule w/o evidence tokens	0.172 $\pm$ 0.111	0.452 $\pm$ 0.066	0.215 $\pm$ 0.039	0.648 $\pm$ 0.052	0.023 $\pm$ 0.010

quadrants and the central disc gives 0.733 (-0.012). With seed-level standard deviations of 0.007–0.022, neither difference can be distinguished from noise. On DDR, then, *the global lesion statistics carry practically all of the evidence that the zone-wise statistics carry*; the zone resolution that motivated the token design (Section 4.11) is not shown to be beneficial. We see two explanations, which we cannot separate with the present data. The quadrants are image quadrants and not anatomically registered ETDRS fields, so they may not capture the spatial pattern that the clinical rules concern; and the grading label of DDR may depend mainly on the amount and type of lesions rather than on their spread, for example because the boundaries between the middle grades are drawn on counts. The result also means that our rule R_3 , which uses quadrant-wise haemorrhage evidence, is less well supported than the other rules.

Lesion classes contribute unequally. Removing haemorrhages costs 0.030 QWK, removing microaneurysms 0.024, removing soft exudates 0.019, and removing hard exudates only 0.004; the last difference is negligible although exudates are the best-segmented class (Section 6.1), presumably because their information is redundant with the other classes and with the image embedding (large exudates are also visible to the ImageNet features). When a single class is kept, microaneurysm evidence reaches 0.704, haemorrhage evidence 0.687, soft-exudate evidence 0.674 and exudate evidence 0.630; the last is no better than the embedding-only models (0.638–0.644; Table 9). It is striking that microaneurysms, the class with the lowest segmentation quality (AUPR at most 0.17), are the most informative *single* class. A plausible reason is

that microaneurysms are the earliest lesion and that ICDR grade 1 is defined by their presence alone, so that even a noisy microaneurysm signal separates grade 0 from higher grades. This implies that better microaneurysm segmentation, which requires higher-resolution input than we could afford, has the greatest potential to improve the grader. As with all ablations in this paper, these are three-seed estimates on a single backbone and should be read as indications.

Table 15: Evidence ablation: the full LesionRule head trained with parts of the evidence tensor removed (zones or lesion classes set to zero everywhere, including in the rule scores). Three seeds (0–2), mean $\pm$ SD, leak-controlled test set. Δ is the mean QWK difference to the reference. With three seeds and SDs of 0.005–0.02, differences below about 0.01 are not distinguishable from noise.

Evidence kept	QWK	Δ QWK	Acc.	ECE
Full evidence (reference)	0.745 $\pm$ 0.007	–	0.748 $\pm$ 0.009	0.048 $\pm$ 0.009
Whole-field zone only	0.736 $\pm$ 0.013	-0.008	0.747 $\pm$ 0.013	0.056 $\pm$ 0.012
Quadrants + centre (no whole field)	0.733 $\pm$ 0.022	-0.012	0.742 $\pm$ 0.010	0.055 $\pm$ 0.024
Without MA	0.721 $\pm$ 0.002	-0.023	0.739 $\pm$ 0.000	0.055 $\pm$ 0.002
Without HE	0.715 $\pm$ 0.011	-0.029	0.729 $\pm$ 0.006	0.058 $\pm$ 0.010
Without EX	0.741 $\pm$ 0.011	-0.003	0.747 $\pm$ 0.003	0.054 $\pm$ 0.021
Without SE	0.726 $\pm$ 0.009	-0.019	0.736 $\pm$ 0.008	0.032 $\pm$ 0.001
MA only	0.704 $\pm$ 0.018	-0.041	0.717 $\pm$ 0.006	0.062 $\pm$ 0.007
HE only	0.687 $\pm$ 0.004	-0.058	0.716 $\pm$ 0.005	0.067 $\pm$ 0.017
EX only	0.630 $\pm$ 0.003	-0.115	0.680 $\pm$ 0.003	0.083 $\pm$ 0.009
SE only	0.674 $\pm$ 0.009	-0.071	0.697 $\pm$ 0.001	0.076 $\pm$ 0.009

6.9. Failure analysis

Fig. 14 shows, for the four most frequent error types of LesionRule, one of the 15 most confident wrong predictions (chosen at random among them with a fixed seed), together with its lesion maps. They illustrate failure modes that the aggregate numbers hide, and we describe them with the caveat that four images do not characterise a distribution.

Moderate called normal (37% of grade-2 images).. The example has a confidence of 0.93 for “normal”, and its lesion maps contain only a few scattered, low-intensity responses (a couple of microaneurysm and haemorrhage cells). In DDR, grade 2 can be assigned on the basis of microaneurysms and a few

small haemorrhages, i.e. on lesions that are at or below the resolution of our segmentation network (512^2 for an image of about 2000 pixels). The confusion is therefore a direct consequence of the weakest component of the pipeline (microaneurysm and small-haemorrhage segmentation, Section 6.1), and it explains why the largest class-wise gains are not to be expected from the grader but from higher-resolution lesion maps.

Normal called moderate.. The example (confidence 0.64) has many exudate responses scattered over the posterior pole. We cannot tell from the image alone whether these are real exudates that were not graded, drusen-like structures or reflection artefacts that the segmentation network mistook for exudates; since the DDR grades were assigned on the full-resolution photograph, each possibility is plausible. This example shows how the evidence-based design transmits segmentation errors directly to the grade, and why an uncertainty-aware output (conformal sets) is meant to help: a moderate-confidence prediction like this one is the situation in which a set with several grades is preferable to a single label.

Severe called moderate.. The grade 3 example has sparse lesion maps. The severe NPDR criteria rely on the extent of haemorrhages in four quadrants, venous beading and intraretinal microvascular abnormalities. Our evidence includes the first only through noisy haemorrhage maps, and has no representation of the other two. With only 61 test images of grade 3 and 118 in training, the head has few examples from which to learn the pattern, and recall is 0.07.

PDR called moderate.. Proliferative DR is defined by neovascularisation, vitreous or pre-retinal haemorrhage, none of which is a segmentation class in DDR. The example shows moderate-looking exudates and a few haemorrhages. This is a structural limitation of any model that uses only the four DDR lesion classes as evidence: the image embedding must provide the missing information, which in our frozen-backbone setting it does only partially (recall of grade 4: 0.58). A segmentation class for neovascularisation, or a fine-tuned backbone, is the natural remedy.

6.10. Comparison with an end-to-end fine-tuned CNN

To place the frozen-embedding results in context, we fine-tuned a ResNet-18 end-to-end on the 6260 gradable training images (Gaussian-filtered, 224×224 crops from 256×256 , ordinal cumulative-link loss with the same soft

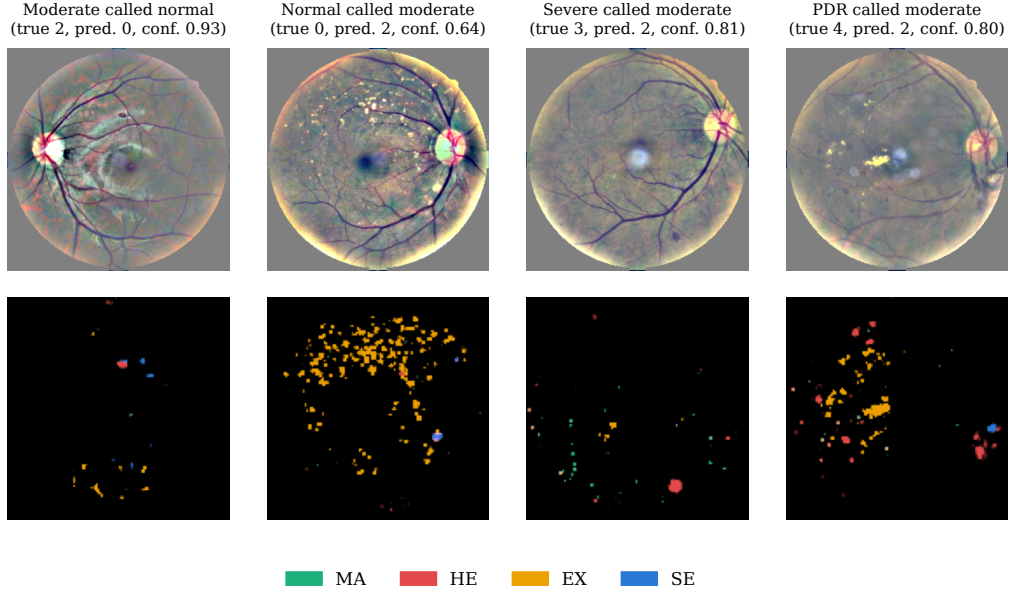


Figure 14: Confidently wrong predictions of LesionRule on the leak-controlled test set (one example per error type, chosen at random among the 15 most confident errors of that type): Gaussian-filtered input (top) and lesion maps (bottom).

targets, AdamW, 8 epochs, one run; about 17 minutes of CPU time). This is a deliberately modest reference rather than a tuned state-of-the-art model: it has a small backbone, low resolution, a short schedule, one seed, and it does not use lesion information.

Table 16 shows that the fine-tuned network reaches a QWK of 0.700, an accuracy of 71.8% and an AUC for referable DR of 0.891, with a low calibration error (0.031). It is clearly better than the frozen-embedding models that do not use lesion evidence (B2: 0.638; paired difference -0.059 for B2 minus fine-tuned, interval [-0.076, -0.042]), which confirms that fine-tuning on DR images adds information that a frozen ImageNet embedding lacks. At the same time, LesionRule remains better than this reference (0.734 versus 0.700; paired difference +0.036, 95% interval [+0.018, +0.053]), although it only uses a frozen backbone, so lesion evidence derived from 383 pixel-annotated images is worth more than end-to-end fine-tuning of a small backbone on 6260 image-level labels in this setting. We do not draw conclusions about larger or longer-trained end-to-end models: a fine-tuned high-resolution network, a stronger backbone or a retinal foundation model would very probably

exceed both, and combining such a backbone with lesion evidence (the natural extension of this work) was beyond our compute. The fine-tuned model shows the same optimism on the validation partition as the other models (validation QWK 0.858 versus 0.700 on the test set), which is consistent with the partition shift documented in Section 6.6.

Table 16: End-to-end fine-tuned CNN reference versus frozen-embedding models on the leak-controlled test set. The fine-tuned model is a single run (trainable parameters in millions); the others are seed averages. Intervals: 95% bootstrap over test images.

Model	Params (M)	QWK [95% CI]	Acc.	F1	AUC _{≥2}	ECE
End-to-end fine-tuned ResNet-18 (1 run, 8 epochs, 224 ²)	11.2	0.700 [0.677, 0.720]	0.718	0.458	0.891	0.031
LesionRule (frozen ConvNeXt-T + evidence, 5 seeds)	0.14 (head)	0.734 [0.714, 0.754]	0.745	0.464	0.923	0.057
B2: frozen ConvNeXt-T, ordinal (5 seeds)	0.30 (head)	0.638 [0.616, 0.662]	0.683	0.428	0.867	0.120

6.11. Computational cost

Table 17 summarises measured costs on the four-core CPU machine. Training the segmentation network for 30 epochs takes about 23 minutes (45 seconds per epoch) with bfloat16 autocast; inference at 512×512 runs at about 0.12 seconds per image (8 images per second) for the network alone and about 3 images per second end-to-end, including decoding of the original 2-megapixel JPEG files, resizing and filtering, which is how the lesion maps for all 13 673 DDR images were produced in roughly 75 minutes. The grading stage is negligible in comparison: the token head has 0.14 M parameters and each of its training runs took a few minutes on a single core (while sharing the machine with other jobs), and evidence extraction for thousands of images takes seconds. The cost of LesionRule over the embedding-only baselines is therefore dominated by one segmentation pass per image. These timings are specific to this machine and are given to show that the whole study is reproducible without a GPU.

Table 17: Model sizes and measured costs (four CPU cores, bfloat16 autocast where applicable).

Component	Role	Parameters	Measured cost
U-Net (ResNet-18 + full-res branch)	lesion segmentation	11.8 M	train 23 min; 0.12 s/image at 5
ConvNeXt-T (frozen)	image embedding	27.8 M	feature extraction, 256^2
MLP baselines (B1, B2, B4)	grading	0.30 M	seconds per run
Evidence MLP (B3)	grading	0.016 M	seconds per run
Evidence-token head (LesionRule)	grading	0.14 M	a few min per run (1 core)

7. Discussion

7.1. What the experiments support

We started from four gaps (Section 1) and proposed a pipeline that couples segmentation, evidence extraction, ordinal grading, rule consistency and conformal referral. The experiments give a mixed but informative picture, which we summarise by component rather than by headline.

Lesion evidence as input to the grader: supported. The clearest result is that lesion-derived statistics carry grading information that a generic image embedding does not capture. A 120-number summary of the segmentation output (B3) is as informative about DR grade as a 1152-dimensional ImageNet embedding (QWK 0.641 vs. 0.638–0.644), and the two are complementary (QWK 0.72–0.73 together, AUC for referable DR 0.92 versus 0.87). The mechanism is plausible and, in our setting, partly an effect of *supervision*: the segmenter is trained on 383 images with pixel-level labels, so the evidence tensor imports expert lesion annotation that the grade labels of 6260 images and a frozen ImageNet backbone cannot provide. We did not isolate this by, for example, training the segmenter with fewer annotated images, so we cannot say how the gain scales with the number of pixel-annotated images; this would be informative for deciding how much annotation effort is justified. That the mild and severe grades remain poorly recognised shows that the coarse lesion statistics are not sufficient for the finest clinical distinctions (a few microaneurysms; the 4–2–1 rule requires venous beading and IRMA, which are not annotated and not segmented).

Zone resolution: not supported. A design premise of the bottleneck was that severity depends on how widely lesions are spread, which should make zone-

wise evidence more informative than global counts. The ablation of Section 6.8 does not support this on DDR: whole-field statistics alone reach QWK 0.736 versus 0.745 with all zones (three seeds, difference within noise). Whether this reflects our crude quadrants, the way DDR labels are assigned, or the quality of the lesion maps cannot be decided here. The same ablation shows that microaneurysm evidence, though from the worst-segmented class, is the most informative single class, whereas hard-exudate evidence is nearly redundant.

Token architecture versus naive fusion: marginal. The transformer head over evidence tokens is only slightly better than concatenation (+0.013 QWK, interval just excluding zero). We would not build a case for the architecture on accuracy. Its justification is that the attention over (c, z) tokens gives a direct view of which lesion class and zone matter for a prediction, which a concatenated MLP does not. We did not evaluate this interpretability quantitatively (for example with a deletion test or with ophthalmologist review), so it remains a design argument rather than a result.

Soft ordinal targets: a calibration effect. Gaussian-smoothed targets reduced the ECE from 0.095 to 0.058 without changing agreement. Since calibration is a precondition for any threshold-based referral or conformal procedure, this is useful, but the effect is on confidence rather than on decisions.

Clinical-rule consistency: not demonstrated. The rule loss did not reduce rule violations in a meaningful way and did not improve accuracy. Our own analysis points to the reason: the data-calibrated necessary conditions are rarely violated by any of the models. We consider this a legitimate negative finding. It does not show that clinical knowledge is useless for DR grading, but it shows that this form of knowledge (soft necessary conditions calibrated on the same labels, with quadrants instead of ETDRS fields, and without the lesions that define severe disease) adds little beyond what the evidence already conveys. The *violation rate* itself remains, we believe, a useful audit measure, and it would become more discriminating with stricter rules.

Conformal referral: valid construction, invalid assumption on DDR partitions. Conformal sets achieved their nominal coverage when calibration and test data were exchangeable and fell short when the calibration set came from the official validation partition. The 8.5-point accuracy difference between the DDR validation and test partitions (84% vs. 75%) is a finding in its own

right: results tuned on the DDR validation set will be optimistic relative to the test set for *any* method, and conformal calibration inherits the optimism. Practitioners should calibrate on data from the deployment distribution, and benchmark papers should report validation-test gaps.

External generalisation: not achieved. The zero-shot transfer to the Gaussian-filtered Kaggle images failed for all models. Because the external images are 224×224 and filtered, the experiment conflates population shift with a resolution mismatch that no method based on tiny-lesion evidence could overcome. It does, however, show that the pipeline as built is tied to the DDR acquisition setting, and it argues against reading the DDR results as evidence of cross-population robustness.

7.2. Relation to prior work

Our results are consistent with the premise of lesion-aware grading approaches (Huang et al., 2021; Sun et al., 2021; Wang et al., 2017; Yang et al., 2017), namely that lesion information helps grading and that image-level labels alone leave this information underused. We do not compare numerically with their published results, because splits, resolutions, backbones and compute differ and we have not re-run their code; absolute numbers from a frozen-backbone, CPU-trained study should not be compared with GPU-trained end-to-end systems. The distinct elements of this study — zone-wise soft evidence as the only route from lesions to grade, rule-based consistency checking, and conformal referral analysed under both exchangeable and shifted calibration — are, to our knowledge, not covered by earlier DR work, though we cannot rule out that related ideas exist in publications that we did not find; the literature comparison in Table 1 is qualitative and based on our reading.

7.3. Clinical interpretation

At the level of a screening programme, the most relevant outputs are the referable-DR sensitivity and specificity. With a hard arg-max decision the lesion-grounded model detects 68% of referable cases at 95% specificity on this test partition. Conformal referral with $\alpha = 0.1$ raises sensitivity to 0.85 under exchangeability at 0.88 specificity, but, because sensitivity of a screening tool should be as high as possible and the partition-shift result shows how fragile such operating points are, any deployment would need local calibration. The poor recognition of mild DR (grade 1) matters less

for referral (mild DR is not referable) than the confusion of moderate with normal (37% of moderate cases), which is where missed referrals arise. We caution against any clinical reading of these retrospective results.

7.4. Practical recommendations

Even with the caveats above, the experiments suggest some recommendations for practitioners who build DR systems from DDR-like data. *First*, if pixel-level lesion annotations exist for even a few hundred images, train a segmenter on them and feed its summary statistics to the grader: this was the largest effect in our study and costs one forward pass per image. A single global summary (counts at a few confidence levels, peaks and maxima per lesion class) was sufficient; zone-wise statistics are not worth the added complexity unless anatomically registered. *Second*, do not use plain cross-entropy for small lesion classes; a Dice or Tversky term was necessary for microaneurysms, and the choice between the two is of second order. *Third*, calibrate and report validation–test gaps: model selection and any probability calibration done on one partition may be optimistic for another, and a split-conformal wrapper is only as valid as the exchangeability of its calibration set. *Fourth*, report internal consistency and failure modes, not only agreement: a few percent of predictions may contradict the lesion evidence, and the dominant error (moderate called normal) is traceable to the weakest component, which tells where to invest. *Fifth*, evaluate across at least one external dataset at native resolution before claiming robustness: our zero-shot transfer failed, and a 224×224 input is unsuitable for lesion-based reasoning.

7.5. Limitations

We list the limitations that we consider material, from most to least consequential for the claims of this paper.

Compute-bound design. All experiments run on four CPU cores. The image embedding is a frozen ImageNet backbone, the segmentation network is trained at 512×512 on 383 images, and we could not train end-to-end grading networks at native resolution, large transformer backbones, retinal foundation models, or heavy detectors. Absolute grading accuracy is therefore lower than what a GPU-trained, fine-tuned system can reach, and the comparison with a fine-tuned ResNet-18 (Section 6.10) is the only end-to-end reference we provide. The *relative* conclusions (evidence tokens versus none, with versus without rule loss) are drawn under identical embeddings, but we

cannot exclude that they change when the backbone is fine-tuned or replaced by a stronger one. This is the most important open question for follow-up work.

Single segmentation run. Segmentation and detection results come from one training run per configuration. Differences of a few thousandths between segmentation variants are within the range that seeds could change; we therefore base claims only on the larger differences, and we explicitly avoid significance statements for segmentation.

Train–test discrepancy in lesion evidence. The lesion maps of the 375 grading-training and validation images that are also lesion-training images (Appendix A) are produced by a network that saw their pixel labels, and are therefore cleaner than those of unseen images. This slightly biases the evidence statistics seen during head training towards better lesion maps than at test time (about 4% of the grading-training images). We removed the corresponding leak from the *test* set, but did not re-train the segmenter on disjoint folds, which would be the cleaner solution.

Rule templates are surrogates. Neovascularisation, intraretinal microvascular abnormalities and venous beading are not annotated in DDR, so rules for severe and proliferative DR are only partial; the zones are image quadrants, not ETDRS fields defined with respect to the optic disc and fovea, because we did not localise those landmarks; and the thresholds are fitted on grade labels, which themselves contain noise. The rules should be read as clinically inspired regularisers and consistency checks, not as an implementation of the grading guideline, and a low violation rate does not by itself prove clinical correctness.

Conformal guarantee. The coverage guarantee is marginal, assumes exchangeability between calibration and test data, and does not hold under distribution shift or conditionally for rare grades (Section 6.6). Because severe DR is rare, class-conditional coverage for grade 3 is estimated from few samples.

External validation. The only external test is a 224×224 , Gaussian-filtered grading set without lesion annotations. It evaluates robustness of the grading pipeline to a shift in population, camera and resolution, but cannot validate lesion segmentation or detection, and it is not a substitute for prospective multi-centre validation. Conformal calibration was not re-done on the external set.

Labels and annotation. DDR grading labels carry inter-grader noise that we model only through soft ordinal targets. Training-set detection boxes contain duplicates (removed), the detection ground truth is derived from the same lesions as the masks and is therefore not an independent annotation, and microaneurysm annotation is known to be inconsistent. Resizing masks to 512×512 with a small area threshold preserves tiny lesions but also dilates some of them by a fraction of a pixel.

No clinical claim. The study is retrospective, on public data. It does not establish clinical utility, safety, or fairness across subgroups; DDR provides no demographic information, so subgroup performance could not be examined.

7.6. Threats to validity

Internal validity. The main threats are selection effects and tuning on the test set. All design decisions of the grading stage (soft-label width, rule weight, architecture sizes, learning rates) were fixed before the test set was evaluated, using validation data and general practice; no test-set-based tuning took place. However, our decisions were taken after we had seen validation behaviour of a small number of configurations, and some were revised after results, as documented in the text (for example, the detection seed protocol was changed from a threshold tied to the Dice-optimal value to a validation-selected value after we observed that the former capped recall, and the leak-controlled test set was introduced after we found that the partitions overlap). Both changes were made before the final test numbers were produced for the affected tables, but we report them so that readers can judge.

Construct validity. QWK and AUC summarise agreement and ranking, but not the clinical consequences of errors. The rule-violation rate is only as meaningful as the rules; the detection metrics depend on box annotations that are derived from masks; and the conformal results depend on the definition of the referral boundary. Likewise, the “leak-controlled” test set is defined by file-name matching between the lesion and the grading partitions; if the same eye appears under different file names, or the same patient with two eyes in different partitions, there may be additional leakage that we cannot detect, because DDR does not provide patient identifiers.

External validity. All conclusions concern one dataset, one segmentation architecture, one image backbone and one resolution. The failure of the

external test shows that they should not be extrapolated to other populations or devices.

Statistical conclusion validity. Five seeds for the grading heads and three for the ablation are few; they quantify optimisation variance but not data variance, which we address with bootstrap intervals over test images. Segmentation was run once per configuration. Multiple comparisons are not corrected in the ablation tables, and we have therefore avoided claims based on differences below the seed-level variability.

7.7. *Ethical considerations and responsible use*

Automated DR screening can extend access to care but can also cause harm when deployed without safeguards: false negatives delay treatment of sight-threatening disease, and systematic performance differences across populations, camera types and image quality can widen disparities. Three features of the proposed framework are meant to support responsible use, although none of them is sufficient on its own: lesion-grounded evidence tokens make the basis for a grade inspectable; the rule-violation rate flags internally inconsistent outputs; and conformal sets quantify when the model should abstain from a single-grade decision and refer the case to a human reader. The models and code are released for research only, in line with the non-commercial licence of DDR; any clinical use would require prospective validation, regulatory review, local calibration, and monitoring for distribution shift.

7.8. *Future work*

Four directions follow directly from the limitations. (i) *End-to-end training*: because the evidence bottleneck is differentiable, the segmenter can be fine-tuned from grade supervision, and the fine-tuned image backbone can be replaced by a retinal foundation model (Zhou et al., 2023); this requires GPUs. (ii) *Weak lesion supervision*: only 5.5% of DDR images (757 of 13,673) have lesion annotations, but almost all have grades. A teacher-student scheme (Tarvainen and Valpola, 2017) with uncertainty gating (Gal and Ghahramani, 2016) and an attention-based multiple-instance loss (Ilse et al., 2018) could use grades as weak lesion supervision; we implemented the building blocks (probability maps, evidence, rules) but did not run this experiment. (iii) *Anatomical zones*: replacing quadrants by ETDRS fields

registered to optic disc and fovea would make the rules closer to the clinical definition. (iv) *Prospective and multi-centre evaluation* with conformal re-calibration under shift.

8. Conclusion

We studied DR grading, lesion segmentation and lesion detection as one coupled problem on the DDR dataset, with the grade computed from zone-wise soft lesion statistics of a segmentation network and complemented by ordinal soft targets, a clinical-rule consistency loss and conformal referral sets. Under a strictly controlled protocol — a leak-controlled test set, five seeds, bootstrap intervals, matched baselines and component ablations — the clearest finding is that lesion evidence adds information that a generic image embedding lacks: the quadratic-weighted kappa rises from 0.64 to 0.73 and the AUC for referable DR from 0.87 to 0.92, with a segmentation network trained on only 383 pixel-annotated images, and the result exceeds a modest end-to-end fine-tuned ResNet-18 reference (0.70). An ablation shows that whole-field lesion statistics carry almost all of this information (zone resolution added nothing detectable), and that microaneurysm and haemorrhage evidence matter most. Soft ordinal targets improve calibration; the token architecture gives a small gain over naive fusion; the rule-consistency loss did not change accuracy or violation rates measurably, because the rules are rarely violated to begin with. Conformal prediction delivers its guarantee under exchangeability but is not robust to the shift we found between the DDR validation and test partitions, and the pipeline does not transfer zero-shot to a low-resolution external set. Segmentation of small lesions (microaneurysms) and detection at strict overlap remain weak at the resolution that our compute allowed, and mild and severe DR are poorly recognised.

These conclusions come with the caveats of a CPU-bound, frozen-backbone study with single segmentation runs. The most valuable next steps are end-to-end fine-tuning with a stronger backbone, using grade labels as weak supervision for the 95% of images without lesion masks, anatomically defined zones and stricter rules, and prospective multi-centre validation with calibration on the deployment distribution. We release code and every intermediate result so that these steps can be taken without re-running the CPU pipeline.

Declarations

CRediT authorship contribution statement. [To be completed by the authors.] Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision.

Declaration of competing interest. [To be completed by the authors.] The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability. The DDR dataset is available from its authors at <https://github.com/nkicsl/DDR-dataset> (links to Google Drive and Baidu Drive; CC BY-NC-SA 4.0). The Gaussian-filtered Kaggle dataset is available at <https://www.kaggle.com/datasets/sovittrath/diabetic-retinopathy-224x224-gaussian-filtered> (CC0). Code, configuration files and result files that generate every table and figure in this paper are available in the repository <https://github.com/deep1789/Diabetic-Retinopathy>.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work an AI assistant (Claude, Anthropic) was used to write code, run experiments and draft the manuscript text. The authors must review and edit all content and take full responsibility for the publication before submission. [Authors to confirm and adapt this statement according to the journal’s policy.]

Ethics. This study uses only publicly available, de-identified images; no new patient data were collected.

Appendix A. Overlap between lesion and grading partitions, and sensitivity analysis

Table A.18 shows how the 757 lesion-annotated images are distributed over the grading partitions, computed by matching file names. The two partitions are not aligned. The 126 grading-test images that also occur in the lesion-training (8) or lesion-validation (118) subsets are removed from the grading test set in all main experiments (leak-controlled test set, 3633

gradable images). Table A.19 compares the main grading metrics on the leak-controlled and on the full official test partition (3759 gradable images).

Table A.18: Number of lesion-subset images (rows) found in each grading partition (columns), by file name.

Lesion subset	Grading train	Grading valid	Grading test	Total
Lesion train (383)	275	100	8	383
Lesion valid (149)	0	31	118	149
Lesion test (225)	152	12	59	223*

* Two lesion-test images are not listed in any grading partition.

Table A.19: Sensitivity to the leak control: QWK (mean $\pm$ SD over 5 seeds) on the leak-controlled test set (3633 images) versus the full official gradable test partition (3759 images).

Model	Leak-controlled	Full official test
B1: embedding, CE	0.644 $\pm$ 0.009	0.641 $\pm$ 0.009
B2: embedding, ordinal	0.638 $\pm$ 0.010	0.637 $\pm$ 0.010
B3: evidence only	0.641 $\pm$ 0.003	0.646 $\pm$ 0.003
B4: embedding + evidence (concat.)	0.722 $\pm$ 0.011	0.723 $\pm$ 0.010
T0: evidence tokens	0.722 $\pm$ 0.014	0.722 $\pm$ 0.014
T1: T0 + soft labels	0.732 $\pm$ 0.014	0.731 $\pm$ 0.014
T2: T0 + rule loss	0.729 $\pm$ 0.017	0.729 $\pm$ 0.017
LesionRule (full)	0.734 $\pm$ 0.009	0.733 $\pm$ 0.009
A1: LesionRule w/o evidence tokens	0.632 $\pm$ 0.015	0.629 $\pm$ 0.016

Appendix B. Hyper-parameters

Table B.20: Hyper-parameters. Values were fixed a priori or tuned on validation data only.

Component	Parameter	Value
Segmentation	input / crop	512 ² full image / 256 ² crops
	optimiser, schedule	AdamW, wd 10 ⁻⁴ , one-cycle (max 10 ⁻³ , 15% warm-up)
	batch, iterations $\times$ epochs	8 crops, 48 $\times$ 30
	positive weights (MA,HE,EX,SE)	20, 6, 6, 8
	focal Tversky	$\alpha = 0.3$, $\beta = 0.7$, $\gamma = 0.75$, weight 2
	lesion-centred crop probability	0.7
Evidence	augmentation	flips, 90° rotations, contrast $\times [0.8, 1.25]$, brightness ± 0.08
	grid, zones	128 ² , 6 (4 quadrants, central disc $r = 0.35R$, whole field)
	soft-count thresholds, temperature	$\tau \in \{0.25, 0.5, 0.75\}$, $T = 0.05$
	peak rule	3 $\times$ 3 local maximum, $\bar{P} > 0.5$
Grade head	token width, layers, heads, FFN	64, 2, 4, 128
	optimiser	AdamW, wd 0.01, cosine, 60 epochs, batch 128
	learning rate	5 $\times$ 10 ⁻⁴ (transformer), 10 ⁻³ (MLPs)
	soft-label width σ_y	0.5
	rule weight λ , rule slope s	1, 0.35
Rules	fitting	Adam (lr 0.05), 400 steps, negative weight 0.5
Calibration	temperature grid	26 values in [0.5, 3]
Conformal	α	0.05, 0.1, 0.2
Detection	seed threshold grid	{0.05, 0.1, 0.2, 0.3, 0.5, 0.7}, chosen per class on validation AP _{0.3}

Appendix C. Properties of the building blocks

Proposition 1 (Soft counts). *For every zone z , $0 \leq n_{c,z}^{(\tau)} \leq |\Omega_z|$; $n_{c,z}^{(\tau)}$ is non-increasing in τ and non-decreasing in every entry of $\tilde{P}_c$; $|\partial n_{c,z}^{(\tau)} / \partial \tilde{P}_c(u)| \leq 1/(4T)$; and as $T \rightarrow 0$ it converges, for all u with $\tilde{P}_c(u) \neq \tau$, to the hard count $|\{u \in \Omega_z : \tilde{P}_c(u) > \tau\}|$.*

Proof. Each summand $\sigma((\tilde{P}_c(u) - \tau)/T)$ lies in $(0, 1)$, is decreasing in τ and increasing in $\tilde{P}_c(u)$, and has derivative $\sigma'(\cdot)/T \leq 1/(4T)$ because $\sigma' \leq 1/4$. Summing over $|\Omega_z|$ terms gives the bounds. As $T \rightarrow 0$, $\sigma((v - \tau)/T) \rightarrow \mathbb{1}[v > \tau]$ for $v \neq \tau$. $\square$

Proposition 2 (Monotone rules and the rule loss). *Let $R_k(Z; \theta)$ be defined by Eqs. (9)–(12). (i) $R_k \in (0, 1)$ and R_k is non-decreasing in every entry $u_{c,z}$ of the evidence, so additional lesion evidence can only relax a rule. (ii) For $\hat{F} \in [0, 1]^4$, $\mathcal{L}_{\text{rule}}(\hat{F}; R) = 0$ if and only if $\hat{F}_k \leq R_k$ for all k . (iii) $\mathcal{L}_{\text{rule}}$ is continuously differentiable in $\hat{F}$ with gradient $\frac{1}{2} \text{ReLU}(\hat{F}_k - R_k)$, which vanishes exactly on consistent predictions, so consistent predictions are not penalised.*

Proof. (i) $a(v; \theta) = \sigma((v - \theta)/s)$ is increasing in v with values in $(0, 1)$. Under the product t-norm, $\prod_i a_i$ is increasing in each a_i , and the dual $1 - \prod_i (1 - a_i)$

is increasing as well; compositions of increasing maps are increasing and remain in $(0, 1)$. (ii) A sum of squares of non-negative terms is zero exactly if every term vanishes. (iii) $\frac{d}{dx}\text{ReLU}(x)^2 = 2\text{ReLU}(x)$, which is continuous and zero for $x \leq 0$; the factor $\frac{1}{4} \cdot 2 = \frac{1}{2}$ follows. $\square$

Proposition 3 (Ordinal soft targets). *For any $\sigma_y > 0$ the targets $F_k^* = \sum_{j \geq k} q_j(y)$ in Eq. (7) satisfy $1 \geq F_1^* \geq \dots \geq F_4^* \geq 0$, and $F_k^* \rightarrow \mathbb{I}[y \geq k]$ as $\sigma_y \rightarrow 0$. Consequently the cumulative-link loss with soft targets has the same minimiser as the hard-label loss when $\sigma_y \rightarrow 0$, and for $\sigma_y > 0$ its population minimiser is $\hat{F}_k = \mathbb{E}[F_k^* \mid x]$, a monotone sequence.*

Proof. $q_j \geq 0$ and $\sum_j q_j = 1$, so tail sums are non-increasing in k and lie in $[0, 1]$. As $\sigma_y \rightarrow 0$, $q_j(y) \rightarrow \mathbb{I}[j = y]$ and hence $F_k^* \rightarrow \mathbb{I}[y \geq k]$. For each k the binary cross-entropy $-F^* \log \hat{F} - (1 - F^*) \log(1 - \hat{F})$ with soft target F^* is minimised in expectation by $\hat{F} = \mathbb{E}[F^* \mid x]$; monotonicity in k follows from monotonicity of F_k^* pointwise. $\square$

Proposition 4 (Split-conformal coverage). *Let $(x_i, y_i)_{i=1}^n$ be calibration data and (x, y) a test pair, all exchangeable, and $\hat{q}_\alpha$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest of the scores $\varsigma_i = 1 - \hat{p}_{y_i}(x_i)$ (set $\hat{q}_\alpha = \infty$ if the index exceeds n). Then $\Pr[y \in \mathcal{C}_\alpha(x)] \geq 1 - \alpha$, and if scores are almost surely distinct, $\Pr[y \in \mathcal{C}_\alpha(x)] \leq 1 - \alpha + 1/(n+1)$.*

Proof. Since $y \in \mathcal{C}_\alpha(x) \iff \varsigma_{n+1} := 1 - \hat{p}_y(x) \leq \hat{q}_\alpha$ and the $n+1$ scores are exchangeable, the rank of ς_{n+1} among them is uniform on $\{1, \dots, n+1\}$ when scores are distinct (and stochastically no smaller otherwise). The event $\varsigma_{n+1} \leq \hat{q}_\alpha$ is the event that its rank is at most $m = \lceil (n+1)(1-\alpha) \rceil$, which has probability $m/(n+1) \geq 1 - \alpha$ and $\leq (1 - \alpha) + 1/(n+1)$. $\square$

Note that the guarantee is marginal over the population (not conditional on the grade), requires the calibration half to be disjoint from model selection and temperature fitting (which is why validation is split into halves A and B), and presumes no distribution shift between calibration and test; Section 6.7 shows what happens under shift.

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